

Vertex crossings in a symmetric Markov multinomial model

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Abstract

We observe a ball bouncing down a Galton board with any number of directions. At each peg it either keeps its direction with some fixed probability or randomly turns to one of the other directions. Its bin measures how often it went each way, a point of a simplex. When the ball rarely turns, the most likely bins are the corners, which only a ball that never turns can reach. We ask when the best bin of each face of the simplex becomes as likely as a corner. To first order every face catches up at the same moment. We break this tie at second order, with an explicit constant for each face. Hence on a long board, as the expected number of turns grows to any fixed multiple of the length's logarithm, the most likely bin jumps once, from the corners straight to the center. A nonuniform start or a weak external field changes the constants, potentially allowing an intermediate face to win.

Keywords: Potts chain; persistent random walk; Markov multinomial distribution; occupation counts; mode; spanning trees.

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1 Introduction

1.1 The question and the answer

Problem 131 of Kagey's Open Problem Collection [1] is about a Galton board on which the ball tends to keep going the same way. At the first peg the ball goes left or right with probability $\frac{1}{2}$, and at every later peg it repeats its previous direction with probability p . Kagey asks when the middle bin is exactly as likely as the two end bins. Our companion paper [2], Paper A below, answers this for the persistent random walk, or Ising chain with free ends. It compares each bin with an explicit Bessel function, with an error bound that holds for every number of steps, and so traps the tie in an explicit interval for every board. Kagey also asks what changes on a tetrahedron. A natural version of that board sends the ball in one of three directions at each peg, and the bins after N steps form a triangle, one layer of the tetrahedron.

Here we let the ball have $d \geq 2$ directions. Its first direction is uniform, and at each later peg it keeps going the same way with probability p (otherwise it turns to one of the other $d - 1$ directions, each with probability $r = (1 - p)/(d - 1)$). After N steps its bin is the vector $n = (n_1, \dots, n_d)$ of the numbers of steps in each direction. So the bins are the points of a simplex, a segment for $d = 2$ and a triangle for $d = 3$. Only a ball that never turns reaches a corner Ne_i , or vertex, so each vertex has probability p^{N-1}/d .

When p is close to 1 the ball rarely turns, and the vertices are the most likely bins. As p decreases the other bins catch up. On the triangle there are two natural candidates to catch up

first, the middle of an edge, where the ball has used two directions, and the middle of the triangle, where it has used all three. We ask which one wins, and when.

Put $L = \log N$ and $T = Nr$. To first order every face of the simplex catches up with the vertices at the same place, namely when T is about L . So the first order cannot answer the question. We break this tie at second order, and the center of the whole simplex wins. For large N , as T grows from 0 to any fixed multiple of L , the most likely bin jumps once, from the vertices straight to the center, and no edge or other proper face holds it in between. On the triangle the middle bin $(\frac{N}{3}, \frac{N}{3}, \frac{N}{3})$ overtakes the corners while the middles of the edges are still about 17% below them. Figure 1 shows this at $N = 24$.

Why every face ties. Take a face with k directions. A ball that uses only these directions must avoid the other $d - k$ directions at every peg, and it does so with probability $(1 - (d - k)r)^{N-1} \approx e^{-(d-k)T}$. So the face carries mass about $\frac{k}{d}e^{-(d-k)T}$, while a vertex carries $\frac{1}{d}e^{-(d-1)T}$. On the face the ball turns about T times toward each direction, so the fractions n_i/N fluctuate by about $1/\sqrt{T}$ around $1/k$. So the mass of the face is spread over about $(N/\sqrt{T})^{k-1}$ bins near its center, and

$$\frac{\text{central bin of the face}}{\text{vertex}} \approx \text{const} \cdot \left(\frac{e^T \sqrt{T}}{N} \right)^{k-1}.$$

The bracket is the same for every face, and it crosses 1 when $T + \frac{1}{2} \log T = L + O(1)$. Only the constant in front can tell the faces apart.

How second order breaks the tie. The constant is $e^{-(k-1)a_k} = k \det(2\pi\Sigma_k)^{-1/2}$, the mass k of the face over the Gaussian volume of the fluctuations, whose covariance is Σ_k/T (Proposition 3.7). So the ratio is $\exp((k-1)[\Psi - a_k])$ with the common bracket $\Psi = T - L + \frac{1}{2} \log T$, which increases as p decreases. Since $a_2 > a_3 > \dots > a_d$, the full face reaches the vertices first, when $\Psi = a_d$, while a face with $k < d$ directions is still below them by the factor $e^{-(k-1)(a_k - a_d)}$. On the triangle this factor is $e^{-(a_2 - a_3)} = 0.827$. After that moment the full center beats every proper face, since $(d-1)(\Psi - a_d) > (k-1)(\Psi - a_k)$ when $\Psi \geq a_d$.

1.2 Setting and notation

Fix $d \geq 2$ and a start law μ on the set $[d] = \{1, \dots, d\}$ of directions, which we also call states. Let $X_1, X_2, \dots$ be the Markov chain with $X_1 \sim \mu$ that keeps its state with probability $p \in [0, 1]$ and moves to each of the other $d - 1$ states with probability $r = (1 - p)/(d - 1)$. Put $\sigma = p - r = (dp - 1)/(d - 1)$. Kagey's board is $d = 2$ with the uniform start.

Bins. The count vector $K_N = (K_{N,1}, \dots, K_{N,d})$ records how often each state occurs among $X_1, \dots, X_N$. A bin is a vector n of nonnegative integers with sum N , and $P_N^\mu(n) = \mathbb{P}(K_N = n)$. For the uniform start $\mu_i = 1/d$ we write $P_N(n)$. The uniform start is the default everywhere except in Section 4.

Faces and vertices. A face is a nonempty set $F \subseteq [d]$, and $\mu(F) = \sum_{a \in F} \mu_a$. A bin n lies on the face $\text{supp } n = \{i : n_i > 0\}$. The vertices are the faces of size 1, that is, the bins Ne_i . A word that reaches Ne_i never switches, so $P_N^\mu(Ne_i) = \mu_i p^{N-1}$, and for the uniform start we write $V_N = p^{N-1}/d$.

Balanced bins. A bin with k positive coordinates is balanced if these coordinates differ by at most 1. For the uniform start all balanced bins with k positive coordinates have the same probability, since they differ only by a permutation of the states. We call it $C_{N,k}(p)$, so $C_{N,1} = V_N$.

Scales. Put $L = \log N$ and $T = Nr$. The expected number of switches (turns) is $(N-1)(1-p) = (d-1)T(1-1/N)$, so T is about the expected number of switches to each other state. Put $\tau = T/L$. For $p > 0$ let $u = r/p$, the odds of a switch to one given state against no switch, and $z = Nu = T/p$. For $p > 1/d$ we have $\sigma > 0$ and also use $v = r/\sigma = u/(1-u)$. Since $z - T = (d-1)Tz/N$, the scales z and T differ by $O(L^2/N)$ when $T = O(L)$. This is smaller than every error term in our asymptotic statements, so these statements hold with either one. Exact statements are written in u .

Crossings. For $2 \leq k \leq d$, the crossing $p_{N,k}$ is the value of p at which $C_{N,k} = V_N$, and $T_{N,k} = N(1 - p_{N,k})/(d-1)$ is the value of T there. We write $p_{N,k}^{(d)}$ when the number of states matters.

The constants. For $k \geq 2$ put

$$a_k = \frac{1}{2} \log(4\pi) - \frac{k + \frac{1}{2}}{k-1} \log k, \quad D_k = e^{-(k-1)a_k} = \frac{k^{k+1/2}}{(4\pi)^{(k-1)/2}}, \quad \Psi_N(T) = T - L + \frac{1}{2} \log T. \quad (1)$$

Numerically $a_2 = -0.4674$, $a_3 = -0.6571$ and $a_4 = -0.8139$.

Matrix form. The transition matrix is $I + (T/N)Q$ with $Q = J - dI$, where J is the all-ones matrix, so Q switches at rate 1 to each other state. The exit rate of a face F with k states is $\lambda_F = d - k$, the principal Dirichlet eigenvalue of $-Q$ on F .

Potts chain. With the uniform start and $0 < p < 1$, a word $x_1 \cdots x_N$ has probability $\frac{1}{d} r^{N-1} (p/r)^{\#\{s < N: x_s = x_{s+1}\}}$. This is the one-dimensional d -state Potts model with free boundaries. For $d = 2$ it is the Ising chain of Paper A.

Asymptotic notation. In each statement the constants in $O(\cdot)$, and the threshold behind “for large N ”, depend only on the quantities declared fixed in that statement (always including d and k). Uniformity in the other variables is stated each time.

1.3 Main results

Theorem 3.1 makes the heuristic of Section 1.1 precise. Uniformly for $cL \leq T \leq CL$, the logarithm of $C_{N,k}/V_N$ is $(k-1)[\Psi_N(T) - a_k]$, up to a next term $-(k-1)/(8T)$ that is again proportional to $k-1$ and an error $O(T^{-2} + T^2/N)$. Part (ii) of that theorem is the first-order statement. With $T = \tau L$, a balanced bin on a face F has probability $\exp(-L[\tau\lambda_F + |F| - 1] + O(\log L))$. At $\tau = 1$ the exponent $\tau\lambda_F + |F| - 1$ equals $d-1$ for every face, so the first order cannot choose.

By Theorem 3.2, for every $N \geq k$ the balanced bin with k positive coordinates crosses the vertices at exactly one $p_{N,k}$, which increases with N , and

$$T_{N,k} + \frac{1}{2} \log T_{N,k} = L + a_k + \frac{1}{8T_{N,k}} + O(L^{-2}). \quad (2)$$

For $d = 2$ the model is the persistent random walk of Paper A. For $k = 2$ equation (2) is the crossing equation of Paper A [2, Proposition 3.8], and $a_2 = \frac{1}{2} \log(\pi/8)$ is its constant c_0 . Since $a_2 > a_3 > \cdots > a_d$, Corollary 3.9 shows that for large N and all $T \leq CL$ the global mode changes exactly once, from the d vertices straight to the balanced bins with d positive coordinates. All the face-center crossings lie in a p -interval of length $(d-1)(a_2 - a_d + o(1))/N$. For small N this can fail. On the triangle with $N = 4$ an edge center can be the mode (Example 3.10).

The same method gives the crossing of every bin whose positive coordinates are of order N (Theorem 3.3). For fractions α its constant is $D(\alpha) = A^2/\sqrt{\det(2\pi\Sigma_\alpha)}$ with $A = \sum_i \sqrt{\alpha_i}$, where Σ_α is the occupation covariance of a tilted chain, and $\det \Sigma_\alpha$ is a weighted count of the spanning

trees of the complete graph K_k (Proposition 3.6). At the center the count is Cayley's k^{k-2} , and $D(\alpha) = D_k$.

Then we perturb the model in two ways. Each acts at order 1, namely the order of the constants a_k . A start law μ changes the constant of a face F against a vertex i to $a_k + \frac{1}{k-1} \log(k\mu_i/\mu(F))$ (Theorem 4.4). The best bin of each face keeps the share $\mu(F)/k$ of the uniform value (Theorem 4.6), so each face gets a line in t and the highest line holds the mode (Corollary 4.7). On the triangle with $\mu = (\frac{1}{2}, \frac{1}{2}, 0)$ this lets an edge hold the mode on a window of width $\log(16/(9\sqrt{3})) = 0.026$ in T (Example 4.9). A Potts field of size h_i/N per visit to state i adds to each face the field averaged over the face (Theorem 5.2), and one field makes every support size win in turn (Corollary 5.4).

Finally we drop the assumption that the positive coordinates are of order N . A Bessel majorant has relative error $O(L^2/N)$ for every positive count vector when $T = O(L)$ (Theorem 6.2). When at least two coordinates are of order N , a small coordinate starts to matter once it has size N/L^2 (Theorem C.1). Near a vertex, with one large count b and small counts of total $o(b)$, the crossing odds are $u = \sqrt{y/b}(1 + o(1))$, where y is the root of a product of Laguerre polynomials (Theorem 6.4).

The paper is organized as follows: Section 2 gives the exact law. Section 3 proves the tie and breaks it. Sections 4 and 5 treat the start law and the field. Section 6 treats bins near the simplex boundary. Section 7 describes the checks and lists open questions. The longer proofs are in the appendices, namely Appendix A for Sections 2 and 3, Appendix B for Sections 4 and 5, and Appendices C and D for Section 6.

1.4 Related work

The law belongs to the theory of Markov multinomial distributions. Edwards [3] introduced the Markov binomial distribution. Let μ be a law on the two outcomes of a trial, success and failure. His trials form a stationary chain with law μ at each trial that moves from i to j with probability $\theta\delta_{ij} + (1 - \theta)\mu_j$. He wrote the generating function of the number of successes as a product of matrices. Wang and Yang extended this chain to finitely many states, and we follow the account of their work in Johnson, Kotz and Balakrishnan [4, Chapter 35, Section 13.2]. The chain of Wang and Yang starts from a law μ and moves from i to j with probability $\theta\delta_{ij} + (1 - \theta)\mu_j$, where $0 \leq \theta \leq 1$, and the generating function of its vector of category counts is a product of matrices. Our chain with the uniform start and $p \geq 1/d$ is the case where μ is uniform and $\theta = \sigma$. So the exact law for the uniform start is a special case of this theory, and we do not claim it as new. Our general start changes only the initial law and keeps the uniform switching, so it is a Wang–Yang chain only when μ is uniform. Hsiau [6, Theorem 1] proves compound Poisson limits for sums over multistate chains with one nearly absorbing state, and Wang and Tang [5] prove Poisson-type limits for additive processes on Markov chains. Whittle [7, formula (8)] gave the exact law of the transition counts of a finite chain. The count vector is determined by the transition counts and the first state, so P_N could also be summed from his formula.

Brydges, van der Hofstad and König [8] give occupation densities for continuous-time chains on the visited faces of the simplex. Huang, Kiouss, Sidoravicius and Tarrés [9, Theorem 1] give the joint density of the local times, the last-exit spanning tree and the net crossing numbers of a walk with symmetric conductances, with Bessel functions. Stadje [10, Theorems 1 and 2] gives the generating functions of the joint numbers of visits of a finite chain. He also proves limit laws when one state becomes nearly absorbing and the chain leaves it with probability of order $1/N$ [10, Theorems 3 and 4]. The other rows of his transition matrix stay fixed, so the counts of the other states stay bounded. Dietz and Sethuraman [11] study chains that switch from i to j at time n

with probability $G(i, j)/n$. They make about $\log N$ switches by time N , but only $O(1)$ of them after time $N/2$, and their occupation fractions have a limit law on the simplex. Our chain makes about $(d-1) \log N$ switches near the crossings, spread evenly in time. Dupuis and Liu [12] give the rate function of the occupation measure, which we compare with our exponent in Section 3. Kassan-Ogly and Filippov [13] solve the Potts chain in a field in the thermodynamic limit. Their paper was not accessible to us through standard means or institutional libraries, but its abstract and introduction are free online, and we use nothing beyond these. Aside from this, no source we could read compares one count vector with a vertex when p depends on N , and none of them finds the face-center crossings or the tie. We take the Bessel and Laguerre formulas from [15].

2 The model and the exact law

We write P_N^μ in three forms. The *run form* counts the switches, and it gives all our exact statements. The *refresh form* uses the transition matrix $\sigma I + rJ$. When $p \geq 1/d$ the chain keeps its state with probability σ and otherwise draws a fresh state uniformly from all d states, possibly the same one. The *ratio form* is the refresh form divided by a vertex (so it measures a bin in units of a corner). For $p > 1/d$ it is a sum of positive terms, and all our asymptotics start from it.

For a positive count vector $n = (n_1, \dots, n_k)$ with total N and a state a , let $W_n^{(a)}(j)$ be the number of words with counts n and j changes that start with a , and let $W_n(j) = \sum_a W_n^{(a)}(j)$. Put

$$S_n^{(a)}(u) = \sum_j W_n^{(a)}(j) u^j, \quad S_n(u) = \sum_a S_n^{(a)}(u) = \sum_j W_n(j) u^j.$$

These polynomials do not depend on d . For a bin n with some zero coordinates, S_n and $S_n^{(a)}$ mean the polynomials of its positive coordinates.

Proposition 2.1 (the generating function). *Put $y_i = x_i t / (1 - \sigma x_i t)$ and $P_0^\mu(0, \dots, 0) = 1$. As formal power series,*

$$\sum_{N \geq 0} \sum_{n_1 + \dots + n_d = N} P_N^\mu(n) x_1^{n_1} \dots x_d^{n_d} t^N = 1 + \frac{\sum_i \mu_i y_i}{1 - r \sum_i y_i}.$$

Proof. Let F_i be the generating function of the words of length at least 1 that end in state i , each weighted by its probability, and let $S = \sum_i F_i$. Conditioning on the last letter gives $F_i = x_i t (\mu_i + p F_i + r(S - F_i)) = x_i t (\mu_i + \sigma F_i + rS)$, that is, $F_i = y_i (\mu_i + rS)$. Summing over i gives $S = \sum_i \mu_i y_i + rS \sum_i y_i$, and the generating function is $1 + S$. $\square$

Proposition 2.2 (the exact law). *Let $N \geq 1$, let n be a bin, and put $F = \text{supp } n$. Sums over m run over the integer vectors with $m_i = 0$ for $i \notin F$ and $1 \leq m_i \leq n_i$ for $i \in F$, and $M = \sum_i m_i$.*

(a) (*refresh form*) With $0^0 = 1$,

$$P_N^\mu(n) = \sum_m \left(\sum_{i \in F} \mu_i m_i \right) \frac{(M-1)!}{\prod_i m_i!} r^{M-1} \sigma^{N-M} \prod_{i \in F} \binom{n_i-1}{m_i-1}.$$

For the uniform start this reads

$$P_N(n) = \frac{1}{d} \sum_m \frac{M!}{\prod_i m_i!} r^{M-1} \sigma^{N-M} \prod_{i \in F} \binom{n_i-1}{m_i-1}.$$

- (b) (run form) If $p > 0$, then $P_N^\mu(n) = p^{N-1} \sum_{a \in F} \mu_a S_n^{(a)}(u)$. In particular $P_N(n)/V_N = S_n(u)$.
- (c) (ratio form) If $1/d < p < 1$, so that $0 < u < 1$ and $v > 0$, then

$$S_n(u) = (1-u)^{N-1} \sum_m M! v^{M-1} \prod_{i \in F} \frac{1}{m_i!} \binom{n_i-1}{m_i-1}, \quad (3)$$

which is a sum of positive terms.

When $\sigma \geq 0$, the refresh form has a direct reading. Cut the word into M blocks at the $M-1$ refreshes. Each refresh lands on a prescribed state with probability r , and each of the other $N-M$ steps keeps the state with probability σ . If m_i blocks carry state i , there are $m_i(M-1)!/\prod_j m_j!$ orders of the block labels that start with i , and $\binom{n_i-1}{m_i-1}$ ways to choose the lengths of the m_i blocks of state i . The proof below does not need $\sigma \geq 0$.

Proof. (a) By the proof of Proposition 2.1, $S = \sum_{M \geq 1} r^{M-1} (\sum_i \mu_i y_i) (\sum_i y_i)^{M-1}$. By the multinomial theorem the coefficient of $\prod_i y_i^{m_i}$ in $(\sum_i \mu_i y_i) (\sum_i y_i)^{M-1}$ is $\sum_i \mu_i m_i (M-1)!/\prod_j m_j!$. With $w_i = x_i t$ we have $y_i = w_i/(1-\sigma w_i)$, and $[w^j](w/(1-\sigma w))^m = \binom{j-1}{m-1} \sigma^{j-m}$ for $j \geq m \geq 1$. For $m=0$ only $j=0$ contributes. Multiplying the coefficients proves (a), and for the uniform start $\sum_i \mu_i m_i = M/d$.

(b) A word that starts with a and has j changes has probability $\mu_a p^{N-1-j} r^j = \mu_a p^{N-1} u^j$. Sum over the words with counts n .

(c) Divide the uniform case of (a) by $V_N = p^{N-1}/d$. Since $r/p = u$ and $\sigma/p = 1-u$, we have $r^{M-1} \sigma^{N-M}/p^{N-1} = u^{M-1} (1-u)^{N-M} = (1-u)^{N-1} v^{M-1}$. $\square$

At $p = 1/d$ we have $\sigma = 0$ and only $M = N$ survives, which gives the multinomial law for the uniform start. At $p = 1$ only $M = 1$ survives, which gives mass μ_i at $N e_i$. For $\sigma < 0$ the terms in (a) can cancel, so for numerical work we then use exact arithmetic or dynamic programming.

2.1 Bessel bounds

For $x \geq 0$ the modified Bessel function of order 1 is $I_1(x) = \sum_{j \geq 0} (x/2)^{2j+1}/(j!(j+1)!)$. Since $\binom{2j+1}{j} \leq 4^j$, comparing coefficients gives $I_1(x) \leq \sinh x \leq e^x$. We also use the expansion [15, Eqs. 10.40.1 and 10.17.1]

$$I_1(x) = \frac{e^x}{\sqrt{2\pi x}} \left(1 - \frac{3}{8x} + O(x^{-2})\right), \quad x \rightarrow \infty,$$

and the function

$$\Phi(x) = \sum_{j \geq 0} \frac{x^j}{j!(j+1)!} = \frac{I_1(2\sqrt{x})}{\sqrt{x}}, \quad \Phi(0) = 1.$$

Since $\binom{2j}{j} \leq 4^j$, we have $1/(j!(j+1)!) \leq 4^j/(2j)!$, and comparing term by term with the even part of the series of $e^{2\sqrt{x}}$ gives $\Phi(x) \leq e^{2\sqrt{x}}$. Lemmas A.1, A.2 and A.3 in Appendix A give the other facts about Φ that we use.

3 The tie and its resolution

In this section the start is uniform. The constants a_k , D_k and Ψ_N are those of (1). Let n be a balanced vector with k positive coordinates and total N . By Proposition 2.2(b), $C_{N,k}/V_N = S_n(u)$, and as a function of u this ratio does not depend on d .

3.1 Every face ties at first order

Theorem 3.1 (every face ties at first order). *Fix $d \geq 2$ and $0 < c < C$.*

(i) *For large N , uniformly for $cL \leq T \leq CL$ and $2 \leq k \leq d$,*

$$\log \frac{C_{N,k}}{V_N} = (k-1)[\Psi_N(T) - a_k] - \frac{k-1}{8T} + O\left(\frac{1}{T^2} + \frac{T^2}{N}\right).$$

(ii) *Put $T = \tau L$. Uniformly for $c \leq \tau \leq C$ and $1 \leq k \leq d$, a face F with k states has*

$$\log C_{N,k} = -L[\tau \lambda_F + k - 1] + O(\log L), \quad \lambda_F = d - k.$$

At $\tau = 1$ the right side is $-(d-1)L + O(\log L)$ for every k .

(iii) *Put $T = L - \frac{1}{2} \log L + t$. Uniformly for t in a compact interval, for each $2 \leq k \leq d$,*

$$\frac{C_{N,k}}{V_N} \longrightarrow e^{(k-1)(t-a_k)}. \quad (4)$$

(iv) $a_2 > a_3 > \dots$.

So the central bin of every face reaches the vertices when $T = L - \frac{1}{2} \log L + O(1)$, and by (iii) the faces differ in this window only through the constants a_k .

Theorem 3.2 (face-center crossings). *For every $2 \leq k \leq d$ and $N \geq k$ there is exactly one $p = p_{N,k} \in (0, 1)$ with $C_{N,k}(p) = V_N(p)$. The balanced bin is more likely than a vertex below $p_{N,k}$ and less likely above it. The crossing $p_{N,k}$ is strictly increasing in N ,*

$$p_{k,k} = \frac{1}{1 + (d-1)(k!)^{-1/(k-1)}},$$

and every crossing has $u \leq (k!)^{-1/(k-1)} < 1$, so $p_{N,k} > 1/d$. For fixed k the value $T_{N,k} = N(1 - p_{N,k})/(d-1)$ of T at the crossing satisfies

$$T_{N,k} + \frac{1}{2} \log T_{N,k} = L + a_k + \frac{1}{8T_{N,k}} + O(L^{-2}),$$

and

$$T_{N,k} = L - \frac{1}{2} \log L + a_k + \frac{\frac{1}{4} \log L - \frac{1}{2} a_k + \frac{1}{8}}{L} + O\left(\frac{(\log L)^2}{L^2}\right). \quad (5)$$

For $d = 2$, $T_{N,2}$ is the crossing $z_N = N(1 - p_N)$ of Paper A [2, Section 3.2], $a_2 = \frac{1}{2} \log(\pi/8)$ is its constant c_0 , and (5) is its third-order expansion [2, Corollary 3.7(c)]. The proofs here do not use the binary case. Figure 1 shows the full-center crossing on the triangle at $N = 24$.

Idea of the proofs. The proofs are in Appendix A. The exact part uses the run form. The polynomial S_n has nonnegative coefficients, and its first nonzero coefficient is $W_n(k-1) = k!$, from the words that use each state in a single run. So S_n increases from 0 to ∞ and crosses 1 exactly once. Its next coefficient $W_n(k) = k!(k-1)(N-k)/2$ grows with N , which moves the root. The asymptotic part uses the ratio form (3). This sum is close to a Bessel integral (Theorem 6.2),

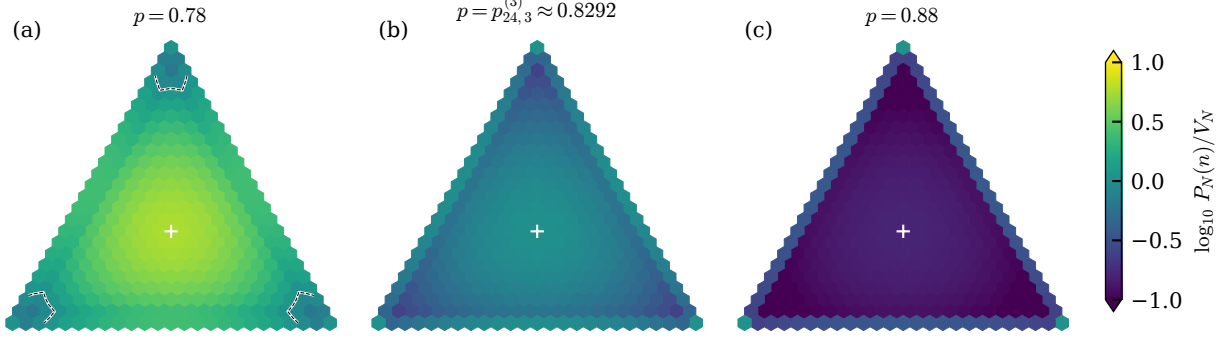


Figure 1: The ratio $P_N(n)/V_N$ for $d = 3$ and $N = 24$, one hexagon per bin, on a $\log_{10}$ scale. (a) At $p = 0.78$ only the 21 bins inside the dashed curves near the vertices are below the vertex height. (b) At the full-center crossing $p_{24,3}^{(3)} = 0.82921 \dots$ the center $(8, 8, 8)$ (cross) ties with the vertices and every other bin is below them. (c) At $p = 0.88$ every nonvertex bin is below the vertex height.

and the Laplace expansion of this integral (Lemma A.4) gives, for a balanced n with k positive coordinates and uniformly for $cL \leq z \leq CL$,

$$S_n(z/N) = D_k \frac{z^{(k-1)/2} e^{(k-1)z}}{N^{k-1}} \left(1 - \frac{k-1}{8z} + O\left(\frac{1}{z^2} + \frac{z^2}{N}\right) \right). \quad (6)$$

This is the heuristic of Section 1.1 with the constants made explicit. Then $\log V_N = -(d-1)T - \log d + O(T^2/N)$ and $z - T = O(L^2/N)$ give parts (i) to (iii) and the expansions of the crossing. Part (iv) follows from the monotonicity of $(x + \frac{1}{2}) \log x / (x - 1)$.

3.2 The constants as Gaussian volumes

By the heuristic of Section 1.1, D_k should be the mass of a face divided by the Gaussian volume of the fluctuations of the fractions. We prove this in two ways, namely at each interior point of a face (Proposition 3.6) and by a sum over the whole face (Proposition 3.7). The next theorem evaluates its constants at the actual fractions $\alpha_i = n_i/N$, since we do not assume that the count vector converges at any rate.

Theorem 3.3 (interior compositions). *Fix $k \geq 2$, $\varepsilon > 0$ and $0 < c < C$. Let $n_1 + \dots + n_k = N$ with $\alpha_i = n_i/N \geq \varepsilon$, and put*

$$A = \sum_{i=1}^k \sqrt{\alpha_i}, \quad g = A^2 - 1, \quad s = k - 1, \quad D(\alpha) = \frac{A^{k/2+1} (\prod_i \alpha_i)^{-3/4}}{(4\pi)^{s/2}}.$$

Uniformly for these count vectors and $cL \leq z \leq CL$,

$$S_n(z/N) = D(\alpha) \frac{z^{s/2} e^{gz}}{N^s} \left(1 + O_{k,\varepsilon,C}(z^{-1} + z^2/N) \right). \quad (7)$$

The unique positive root u_n of $S_n(u) = 1$ satisfies

$$Nu_n = \frac{s}{g} L - \frac{s}{2g} \log L - \frac{\log D(\alpha) + (s/2) \log(s/g)}{g} + O_{k,\varepsilon} \left(\frac{\log L}{L} + \frac{L^2}{N} \right). \quad (8)$$

For a fixed target α and counts $n_i = N\alpha_i + O(1)$, the constants may be evaluated at the target without changing the error order.

So a bin with fractions α grows like $e^{g(\alpha)z}$ against a vertex and crosses at $z \approx (s/g)L$. By Cauchy–Schwarz $g \leq k-1$, with equality only at the center, where $D(\alpha) = D_k$. Since $z - T = O(L^2/N)$, both displays also hold with T in place of z and of Nu_n .

The exponent g is a drop in a rate function (the cost of holding the fractions near α). By Dupuis and Liu [12, Eq. (3.2)], the occupation measure of the chain with generator $Q = J - dI$ has rate function $I(\alpha) = -\langle \sqrt{\alpha}, Q\sqrt{\alpha} \rangle = d - (\sum_i \sqrt{\alpha_i})^2$. So $g(\alpha) = I(e_1) - I(\alpha)$, the drop in the rate from a vertex to α . The next lemma computes the determinant of an occupation covariance for any reversible chain.

Lemma 3.4 (occupation covariance). *Let G be an irreducible generator on $\{1, \dots, k\}$, reversible with respect to $\varpi > 0$, with conductances $c_{ij} = \varpi_i G_{ij} = c_{ji}$ for $i \neq j$. For $\theta \in \mathbb{R}^k$ let $f = \theta - \varpi(\theta)\mathbf{1}$, and let χ solve $-G\chi = f$ with $\varpi(\chi) = 0$. Let Σ be the matrix of the quadratic form $\theta \mapsto 2\sum_i \varpi_i f_i \chi_i$ on the vectors with $\theta_1 = 0$, in the coordinates $\theta_2, \dots, \theta_k$. Let $\mathcal{L}_{(1)}$ be the Laplacian $\mathcal{L} = -\text{diag}(\varpi)G$ without its first row and column. Then*

$$\det \Sigma = \frac{2^{k-1}(\prod_i \varpi_i)^2}{\det \mathcal{L}_{(1)}} = \frac{2^{k-1}(\prod_i \varpi_i)^2}{\text{tree}(c)}, \quad \text{tree}(c) = \sum_{\mathcal{T}} \prod_{\{i,j\} \in \mathcal{T}} c_{ij},$$

where the sum is over the spanning trees $\mathcal{T}$ of the complete graph on $\{1, \dots, k\}$.

The form $2\sum_i \varpi_i f_i \chi_i$ is the usual asymptotic variance per unit time of $\int_0^t \theta(Y_s) ds$ for the chain Y with generator G , but we use it only as a definition. The proof is in Appendix A. For product conductances the tree sum has a closed form. This is classical, and we include a short proof.

Lemma 3.5 (weighted Cayley formula). *For $x_1, \dots, x_k > 0$ and conductances $c_{ij} = x_i x_j$,*

$$\text{tree}(c) = x_1 x_2 \cdots x_k (x_1 + \cdots + x_k)^{k-2}.$$

With every $x_i = 1$ this is Cayley’s count k^{k-2} of the spanning trees of K_k .

Proof. Put $X = \sum_i x_i$. The Laplacian is $\text{diag}(Xx_i) - xx^\top$, so $\mathcal{L}_{(1)} = \text{diag}(Xx_2, \dots, Xx_k) - x'x'^\top$ with $x' = (x_2, \dots, x_k)$. The matrix determinant lemma gives $\det \mathcal{L}_{(1)} = X^{k-1} \prod_{i \geq 2} x_i (1 - \sum_{i \geq 2} x_i/X) = X^{k-2} \prod_i x_i$, and this is $\text{tree}(c)$ by the matrix-tree theorem. For integer conductances this is the multigraph case [14], and both sides are polynomials in the c_{ij} . $\square$

Proposition 3.6 (the tree form of the constants). *Fix $k \geq 2$ and α in the open simplex. Put $\phi = \sqrt{\alpha}$ entrywise and $A = \sum_i \phi_i$.*

(1) *The chain G^α with rates $G_{ij}^\alpha = \phi_j/\phi_i$ for $i \neq j$ is reversible with stationary law α and conductances $\phi_i \phi_j$. Let $Q_k = J - kI$ and $\theta_\alpha = -(Q_k \phi)/\phi$ entrywise. Then $Q_k + \text{diag}(\theta_\alpha)$ is symmetric with Perron root 0 and Perron vector ϕ , and G^α is its Doob transform $\text{diag}(\phi)^{-1}(Q_k + \text{diag}(\theta_\alpha))\text{diag}(\phi)$. It is not a Doob transform of Q_k itself unless α is uniform.*

(2) $\text{tree}(\alpha) := \sum_{\mathcal{T}} \prod_{\{i,j\} \in \mathcal{T}} \sqrt{\alpha_i \alpha_j} = (\prod_i \sqrt{\alpha_i}) A^{k-2}$.

(3) *With Σ_α the occupation covariance of G^α from Lemma 3.4, the constant of Theorem 3.3 is*

$$D(\alpha) = \frac{A^2}{(4\pi)^{(k-1)/2}} \frac{\sqrt{\text{tree}(\alpha)}}{\prod_i \alpha_i} = \frac{A^2}{\sqrt{\det(2\pi \Sigma_\alpha)}}.$$

- (4) At the center $G^\alpha = Q_k$, each of the k^{k-2} spanning trees has weight k^{1-k} , and $\text{tree}(\alpha) = 1/k$. In the coordinates $\alpha_2, \dots, \alpha_k$, with I and J of size $k-1$, $\Sigma_k = (2/k^2)(I - J/k)$ and $\det \Sigma_k = 2^{k-1} k^{1-2k}$. So

$$D_k = \frac{k \cdot k^k}{(4\pi)^{(k-1)/2}} \sqrt{k^{k-2} \cdot k^{1-k}} = e^{-(k-1)a_k}.$$

- (5) The same holds for the constant $D_{k_1}(\beta)$ of Theorem C.1, with k_1 and β in place of k and α .

The proof is a direct check, given in Appendix A. The second way sums the whole face. Its total mass is explicit, and comparing it with the Gaussian sum of the interior profile gives D_k again.

Proposition 3.7 (mass balance). *Fix $k \geq 2$, and let the sums run over the positive vectors $n \in \mathbb{Z}^k$ with $n_1 + \dots + n_k = N$.*

- (1) *For every $N \geq k$ and every u ,*

$$\sum_n S_n(u) = \sum_{j=1}^k (-1)^{k-j} \binom{k}{j} j (1 + (j-1)u)^{N-1}.$$

- (2) *For fixed $0 < c < C$, uniformly for $cL \leq z \leq CL$ and $u = z/N$, this sum is $k e^{(k-1)z} (1 + o(1))$, and it is also $D_k (2\pi)^{(k-1)/2} (\det \Sigma_k)^{1/2} e^{(k-1)z} (1 + o(1))$.*

- (3) *Hence $D_k = k \det(2\pi \Sigma_k)^{-1/2}$.*

Proof of (1). A word of length N over a set U of letters with exactly j changes is fixed by its first letter, the positions of the changes and the new letter at each change. So there are $|U|(|U|-1)^j \binom{N-1}{j}$ of them. By inclusion and exclusion over the set $U \subseteq \{1, \dots, k\}$ of allowed letters, the number of such words that use all k letters is $\sum_U (-1)^{k-|U|} |U|(|U|-1)^j \binom{N-1}{j}$, and this is $\sum_n W_n(j)$. Multiply by u^j and sum over j . $\square$

Parts (2) and (3) are proved in Appendix A. The term $j = k$ in (1) is $P_N(\text{supp } K_N \subseteq F)/V_N$ for a face F with k states. So D_k is the mass k of the face, in units of a vertex, divided by the Gaussian volume $\det(2\pi \Sigma_k)^{1/2}$ of the occupation fluctuations. This is an interpretation, not an independent derivation of a_k , since part (2) uses Theorem 3.3 near the center.

3.3 Balanced bins and the global mode

So far we have compared balanced bins with vertices. To find the global mode we also need that the balanced bins are the best bins of their face.

Theorem 3.8 (balancing within a support). *For $1/d \leq p < 1$, among bins with a prescribed set of positive coordinates and total N , the probability is largest exactly at the balanced bins. More precisely, moving one unit from a coordinate b to another coordinate a , where $b \geq a + 2$ and $a \geq 1$, strictly increases the probability of the bin.*

Once the number of runs of each state is fixed, splitting the visits evenly gives the most ways to choose the run lengths. In the proof in Appendix A this becomes the log-concavity in n of the polynomials B_n of (17). The condition $p \geq 1/d$ holds at every crossing, by Theorem 3.2.

Corollary 3.9 (exactly one mode change). *Fix $d \geq 2$ and $C > 1$. There is N_0 such that for $N \geq N_0$ and every p with $0 < T \leq CL$ the following hold.*

- (i) *If $p > p_{N,d}$, the global modes are exactly the d vertices.*
- (ii) *If $p < p_{N,d}$, they are exactly the balanced bins with d positive coordinates.*
- (iii) *If $p = p_{N,d}$, they are these two classes and nothing else.*

Moreover $p_{N,2} < p_{N,3} < \dots < p_{N,d}$ and $p_{N,d} - p_{N,k} = (d-1)(a_k - a_d + o(1))/N$. So all face-center crossings lie in a p -interval of length $(d-1)(a_2 - a_d + o(1))/N$, and as p decreases the full center crosses first.

By Theorem 3.8 every global mode is a vertex or a balanced bin, and Theorem 3.1(i) compares these as in Section 1.1. The proof is in Appendix A.

Example 3.10 (an intermediate face can be a finite-size mode). Corollary 3.9 does not hold for every N . Take $d = 3$, $N = 4$ and $p = 10/17$, so $u = 7/20$. For an edge center and a full center,

$$\frac{P_4(2, 2, 0)}{V_4} = 2u + 2u^2 + 2u^3 = \frac{4123}{4000} > 1, \quad \frac{P_4(2, 1, 1)}{V_4} = 6u^2 + 6u^3 = \frac{3969}{4000} < 1.$$

Here $p > 1/3$, so by Theorem 3.8 the three edge centers are the only global modes. Also $0 < T = 14/17 < \log 4$, so this p lies in the range of Corollary 3.9 for every $C > 1$. Hence $N_0 > 4$ when $d = 3$.

4 A general start

Suppose now that the ball starts in state a with probability μ_a , and moves as before. The start affects only the first run (the steps before the first turn), but it changes the constants a_k by an amount of order 1, the order at which the faces are separated. So a start can decide which face wins.

For a count vector n with support F , Proposition 2.2(b) gives

$$P_N^\mu(n) = p^{N-1} \sum_{a \in F} \mu_a S_n^{(a)}(u) = p^{N-1} S_n(u) \sum_{a \in F} \mu_a \omega_a(n), \quad \omega_a(n) = \frac{S_n^{(a)}(u)}{S_n(u)}.$$

So the start enters only through the weights $\omega_a(n)$, which sum to 1. A vertex has $V_i^\mu = \mu_i p^{N-1}$. Throughout this section, F is a face with $k = |F| \geq 2$ states and $\mu(F) > 0$ (if $\mu(F) = 0$, every bin with support F has probability 0), and i is a vertex with $\mu_i > 0$.

Lemma 4.1 (start weights). *Let n be a positive count vector on F .*

- (1) *If $n_a = n_b$, then $\omega_a(n) = \omega_b(n)$. So $\omega_a(n) = 1/k$ for all $a \in F$ when n is balanced and k divides N .*
- (2) *Fix $C > 0$. There are C_1 and N_0 , depending only on d , k and C , such that for $N \geq N_0$, $0 < T \leq CL$ and every balanced bin n of F ,*

$$\left| \sum_{a \in F} \mu_a \omega_a(n) - \frac{\mu(F)}{k} \right| \leq \mu(F) C_1 \frac{(1+T)^2}{N}.$$

So a balanced bin forgets where the path started, up to a relative error $O(T^2/N)$. The proof is in Appendix B. For other bins we look at the first run of the path.

Lemma 4.2 (the first run). *Let n be a positive count vector on F and let $a \in F$. Then, as polynomials in u ,*

$$S_n^{(a)}(u) = u \sum_{j=1}^{n_a} (1-u)^{j-1} S_{n-j e_a}(u).$$

Here $S_{n-n_a e_a}$ is the polynomial of the $k-1$ positive coordinates of $n-n_a e_a$, and it equals 1 when $k=2$.

Proof. A word $x_1 \cdots x_N$ with i changes has weight u^i in S_n . Note that $u^i = \prod_{s \leq N} c(x_s, x_{s+1})$ with $c(x, y) = (1-u)\mathbf{1}\{x=y\} + u$, since $c(x, x) = 1$ and $c(x, y) = u$ for $x \neq y$. Expand every factor into its two terms. The term with no u is nonzero only for a constant word, and a word that uses $k \geq 2$ states is not constant. For a word that starts with a and a nonzero term of its expansion, let j be the first step at which the term u is chosen. The steps before j give $(1-u)^{j-1}\mathbf{1}\{x_1 = \cdots = x_j = a\}$, so $x_1 = \cdots = x_j = a$ and $j \leq n_a$. Step j gives u , whatever x_{j+1} is. The later factors, summed over their terms, give the weight of the word $x_{j+1} \cdots x_N$, which can be any word with counts $n - j e_a$. Summing over these words gives $S_{n-j e_a}(u)$. $\square$

In the refresh reading of Section 2, the path stays in a until its first refresh, and after it the path no longer depends on where it started.

Proposition 4.3 (where a conditioned path starts). *Fix $\varepsilon > 0$ and $0 < c < C$, and put $\alpha = n/N$ and $A = \sum_{b \in F} \sqrt{\alpha_b}$. Uniformly for positive count vectors n on F with every $\alpha_b \geq \varepsilon$, for $a \in F$ and for $cL \leq T \leq CL$,*

$$\omega_a(n) = \frac{\sqrt{\alpha_a}}{A} + O\left(\frac{1}{L}\right).$$

Under the uniform start, $\omega_a(n)$ is the probability that the chain starts in a given $K_N = n$. So a path conditioned to have fractions α starts in a with probability close to $\sqrt{\alpha_a}/A$ rather than α_a . Note that $\sqrt{\alpha}$ is the Perron vector ϕ of Proposition 3.6. The proof is in Appendix B. By Lemma 4.2 and Theorem 3.3, removing j visits from a lowers S_n by a factor of about $e^{-ju(A/\sqrt{\alpha_a}-1)}$, and the sum over j gives $\sqrt{\alpha_a}/A$. For $d=2$, $\alpha_1=0.6$, $z=L-\frac{1}{2}\log L$ and $N=2000$ the error is -0.0020 , about $-0.015/L$ (numerically verified, see Section 7).

Theorem 4.4 (face-center crossings with a start). *Let n be a balanced bin of F .*

- (1) *For every $N \geq k$, the ratio $P_N^\mu(n)/V_i^\mu$ is a polynomial in u with nonnegative coefficients, zero constant term and lowest term $(\mu(F)/\mu_i)(k-1)!u^{k-1}$. So it equals 1 at exactly one $p \in (0, 1)$, and the bin is more likely than the vertex below this p . If k divides N , then $P_N^\mu(n) = \frac{\mu(F)}{k} p^{N-1} S_n(u) = \frac{d\mu(F)}{k} P_N(n)$. So the crossing solves $S_n(u) = k\mu_i/\mu(F)$, its u decreases strictly along $N \in k\mathbb{Z}$, and at $N=k$ it is $u = (\mu_i/(\mu(F)(k-1)!))^{1/(k-1)}$.*
- (2) *Fix $C > 0$ and let N_0 be as in Lemma 4.1(2). For $N \geq N_0$, uniformly for $0 < T \leq CL$ and over all μ , $P_N^\mu(n) = \frac{\mu(F)}{k} p^{N-1} S_n(u) (1 + O((1+T)^2/N))$.*
- (3) *Put $a_k(\mu) = a_k + \frac{1}{k-1} \log \frac{k\mu_i}{\mu(F)}$. Fix $0 < c < C$. For large N , uniformly for $cL \leq T \leq CL$ and over all μ ,*

$$\log \frac{P_N^\mu(n)}{V_i^\mu} = (k-1) [\Psi_N(T) - a_k(\mu)] - \frac{k-1}{8T} + O\left(\frac{1}{T^2} + \frac{T^2}{N}\right).$$

Fix also $\Delta > 0$. Uniformly over the μ with $|\log(k\mu_i/\mu(F))| \leq \Delta$, the value $T = T_N^\mu(n)$ of T at the crossing of (1) satisfies $T + \frac{1}{2} \log T = L + a_k(\mu) + \frac{1}{8T} + O(L^{-2})$ and the expansion (5) with $a_k(\mu)$ in place of a_k , and with $T = L - \frac{1}{2} \log L + t$ the ratio tends to $e^{(k-1)(t-a_k(\mu))}$, uniformly for t in compact sets. Here the constants depend also on Δ .

Note that a start changes only the constant of each crossing. It lowers a_k by $\frac{1}{k-1} \log(\mu(F)/(k\mu_i))$, the logarithm of the average start weight on the face against the start weight of the vertex, shared among the $k-1$ new states. When k does not divide N the crossing can depend on the balanced bin n , which is why we write $T_N^\mu(n)$.

Proof. (1) Each $S_n^{(a)}$ has nonnegative coefficients, zero constant term and lowest term $(k-1)! u^{k-1}$, from the $(k-1)!$ orders of the states that start with a , with each state in a single run. So the ratio increases strictly from 0 to ∞ in u , and $u = r/p$ decreases from ∞ to 0 on $(0, 1)$. If k divides N , Lemma 4.1(1) gives $\sum_a \mu_a \omega_a(n) = \mu(F)/k$. The rest of (1) is about S_n , which increases strictly along $N \in k\mathbb{Z}$ at every $u > 0$ and equals $k! u^{k-1}$ at $N = k$, by the proof of the exact part of Theorem 3.2.

(2) is Lemma 4.1(2), divided by $\mu(F)/k$.

(3) By (2), $\log(P_N^\mu(n)/V_i^\mu) = \log S_n(u) - \log(k\mu_i/\mu(F)) + O(T^2/N)$, and $S_n(u) = C_{N,k}/V_N$. So Theorem 3.1(i) gives the display. The rest follows as in Step 3 of the proof of the asymptotic assertions of Theorem 3.2, with $a_k(\mu)$ in place of a_k , since $|a_k(\mu) - a_k| \leq \Delta$ and the ratio increases strictly in T by (1). $\square$

If μ is constant on F , say $\mu_a = m$ for $a \in F$, then $P_N^\mu(n) = m p^{N-1} S_n(u)$ for every n with support in F . So the statements of Sections 3 and 6 about such bins hold verbatim, with V_N replaced by $V_a^\mu = m p^{N-1}$ for a vertex $a \in F$. In particular Theorem 3.8 holds on F . Otherwise balancing can fail.

Example 4.5 (balancing can fail). Let $d = 2$, $N = 4$ and $\mu = (\frac{1}{2} + \varepsilon, \frac{1}{2} - \varepsilon)$ with $0 < \varepsilon \leq \frac{1}{2}$. Counting the changes in the four words with counts (3, 1) and the six with counts (2, 2) gives $P_4^\mu(3, 1) = p^3(u + 2\mu_1 u^2)$ and $P_4^\mu(2, 2) = p^3(u + u^2 + u^3)$. So $P_4^\mu(3, 1) - P_4^\mu(2, 2) = p^3 u^2 (2\varepsilon - u)$, and the unbalanced bin is more likely when $u < 2\varepsilon$.

By Theorem 4.4(2), a balanced bin of F has the share $\mu(F)/k$ of $dC_{N,k}$, up to a factor $1 + O(T^2/N)$. The next theorem shows that no bin of F , balanced or not, does better by more than a factor $1 + O(1/L)$.

Theorem 4.6 (face maxima with a start). *Let F be a face with $k \geq 2$ states.*

(1) *Fix $0 < c < C$. For large N , uniformly for $cL \leq T \leq CL$ and over all μ with $\mu(F) > 0$,*

$$\max_{\text{supp } n=F} P_N^\mu(n) = \frac{d\mu(F)}{k} C_{N,k} \left(1 + O\left(\frac{1}{L}\right)\right).$$

So with $T = L - \frac{1}{2} \log L + t$, uniformly for t in compact sets and over all such μ ,

$$\log \frac{\max_{\text{supp } n=F} P_N^\mu(n)}{p^{N-1}} = \log \frac{\mu(F)}{k} + (k-1)(t - a_k) + o(1).$$

(2) *For n with support F put $y = \sqrt{L}(n/N - \frac{1}{k}\mathbf{1})$. Fix a compact set $\mathcal{J}$ of t and $R > 0$. For large N , uniformly for $t \in \mathcal{J}$ and over all μ with $\mu(F) > 0$, every bin that maximizes P_N^μ among the bins with support F has $|y| \leq R$.*

So a start can move the best bin of F away from the balanced bins, but it gains at most a factor $1 + O(1/L)$, and in the window the best bin stays within $o(L^{-1/2})$ of the center. The bound holds for every bin of F , including bins with coordinates that are $o(N)$. The proof is in Appendix B, and it rests on Lemma 4.2. By Theorem 3.8, each $S_{n-je_a}(u)$ is at most the balanced value at total $N - j$, which is about $e^{-(k-1)ju}$ times the balanced value at total N . So $S_n^{(a)}(u)$ is at most about $u \sum_j (1-u)^{j-1} e^{-(k-1)ju} \approx 1/k$ times the balanced value, and $1/k$ is the start weight of each state at a balanced bin.

Each face now gets a line in the window coordinate t , and the highest line wins.

Corollary 4.7 (global modes with a start). *Order the start weights as $\mu_{(1)} \geq \dots \geq \mu_{(d)}$, put $M_k = \mu_{(1)} + \dots + \mu_{(k)}$ and $T = L - \frac{1}{2} \log L + t$, and define the lines*

$$\ell_1^\mu(t) = \log \mu_{(1)}, \quad \ell_k^\mu(t) = \log \frac{M_k}{k} + (k-1)(t - a_k) \quad (2 \leq k \leq d).$$

Fix a compact set $\mathcal{J}$ of t on which $\max_k \ell_k^\mu(t)$ is attained only at $k = k^$. For large N and all $t \in \mathcal{J}$, every global mode of P_N^μ has exactly k^* positive coordinates, and its support S has $\mu(S) = M_{k^*}$. If $k^* = 1$, the global modes are exactly the vertices i with $\mu_i = \mu_{(1)}$. If $k^* \geq 2$ and μ is constant on such a support S , every global mode with support S is a balanced bin, and all balanced bins of S have the same probability. If $k^* \geq 2$, then for every fixed $R > 0$ and large N , every global mode n has $|n_i/N - 1/k^*| \leq R/\sqrt{L}$ for each i in its support.*

The proof is in Appendix B.

Remark 4.8 (the order of the winners). The winning size is nondecreasing in t . Indeed, if k_1 attains $\max_k \ell_k^\mu$ at t_1 and k_2 attains it at $t_2 > t_1$, adding the two inequalities $\ell_{k_1}^\mu(t_1) \geq \ell_{k_2}^\mu(t_1)$ and $\ell_{k_2}^\mu(t_2) \geq \ell_{k_1}^\mu(t_2)$ gives $(k_2 - k_1)(t_2 - t_1) \geq 0$. Put $t^* = a_d + \frac{1}{d-1} \log(d\mu_{(1)})$, where ℓ_1^μ and ℓ_d^μ cross. Since the slope $k-1$ of ℓ_k^μ lies strictly between the slopes 0 and $d-1$, a size $2 \leq k \leq d-1$ can be the unique winner at some t only if $\ell_k^\mu(t^*) > \ell_1^\mu(t^*)$. For the uniform start $t^* = a_d$ and $\ell_k^\mu(t^*) - \ell_1^\mu(t^*) = (k-1)(a_d - a_k) < 0$, in line with Corollary 3.9.

Example 4.9 (the start decides which face wins). Let $d = 3$ and $\mu = (\frac{1}{2}, \frac{1}{2}, 0)$, and put $t_{\text{ef}} = \log \frac{3}{2} + 2a_3 - a_2 = \frac{1}{2} \log \frac{32\pi}{243}$. Fix t and let N be large. If $t < a_2$, the global modes are $(N, 0, 0)$ and $(0, N, 0)$. If $a_2 < t < t_{\text{ef}}$, they are exactly the balanced bins of the edge $\{1, 2\}$. If $t > t_{\text{ef}}$, every global mode has three positive coordinates. The window $(a_2, t_{\text{ef}}) = (-0.4673558, -0.4412978)$ has width $\log \frac{16}{9\sqrt{3}} = 0.0260580$, while with the uniform start no proper face wins (Corollary 3.9).

Proof. Here $M_1 = \frac{1}{2}$ and $M_2 = M_3 = 1$, so the lines of Corollary 4.7 are $\ell_1^\mu = \log \frac{1}{2}$, $\ell_2^\mu = \log \frac{1}{2} + t - a_2$ and $\ell_3^\mu = \log \frac{1}{3} + 2(t - a_3)$. Hence $\ell_2^\mu > \ell_1^\mu$ exactly when $t > a_2$, $\ell_2^\mu > \ell_3^\mu$ exactly when $t < t_{\text{ef}}$, and $\ell_3^\mu > \ell_1^\mu$ exactly when $t > t^*$, where $t^* = a_3 + \frac{1}{2} \log \frac{3}{2}$, which is the point t^* of Remark 4.8. By (1), $t_{\text{ef}} - a_2 = \log \frac{3}{2} + 2(a_3 - a_2) = 4 \log 2 - \frac{5}{2} \log 3 = \log \frac{16}{9\sqrt{3}}$. This is positive, since $16^2 = 256 > 243 = (9\sqrt{3})^2$. Also $t^* - a_2 = \frac{1}{2}(t_{\text{ef}} - a_2)$. So $a_2 < t^* < t_{\text{ef}}$, and the unique winning size is 1 for $t < a_2$, 2 for $a_2 < t < t_{\text{ef}}$ and 3 for $t > t_{\text{ef}}$. Now apply Corollary 4.7 with $\mathcal{J} = \{t\}$. The vertices with $\mu_i = \frac{1}{2}$ are 1 and 2. The only edge with $\mu(S) = 1$ is $\{1, 2\}$, and μ is constant on it. So for $a_2 < t < t_{\text{ef}}$ every global mode is a balanced bin of $\{1, 2\}$. Hence the global maximum equals the maximum over the bins with support $\{1, 2\}$, and since all balanced bins of this edge have the same probability, each of them attains it. $\square$

Dynamic programs over all bins at $N = 90, 300$ and 1002 show the three phases at finite N , with the window moved to larger t (numerically verified, see Section 7). The next remark explains the shift.

Remark 4.10 (the window at finite N). Let u_N be the unique positive root of $S_{(\lceil N/2 \rceil, \lfloor N/2 \rfloor)}(u) = 1$, which is the crossing odds $(1 - p_N)/p_N$ of Paper A [2, Theorem 3.1]. Since μ is constant on $\{1, 2\}$, we have $P_N^\mu(n_1, n_2, 0)/P_N^\mu(N, 0, 0) = S_{(n_1, n_2)}(u)$, so for every $N \geq 2$ the balanced bins of the edge cross the vertices exactly at $u = u_N$, that is, at $T = Nu_N/(1 + 2u_N)$. This differs from Paper A's $z_N = Nu_N/(1 + u_N)$ by $O(L^2/N)$. Near $t = a_2$ the lines of the full face and of the edges through 3 stay below those of the vertices, so for large N this crossing is the lower end of the window. For the two ends at finite N we use $z = Nu = T/p$. A heuristic calculation, which we do not prove, puts them at $z - L + \frac{1}{2} \log z = a_2 + \frac{1}{8z} + \frac{z^2}{2N} + \dots$ and $t_{\text{ef}} + \frac{1}{12z} + \frac{3z^2}{2N} + \dots$. Only $a_2 + \frac{1}{8z}$ is proved, by Theorem 3.2. The terms z^2/N come from a discrete term $-\frac{(k-1)^2}{2} \frac{z^2}{N}$ in the logarithm of (6). Exact computations give the widths 0.0843, 0.0324, 0.0223 and 0.0214 at $N = 300, 3000, 30000$ and 300000, and the heuristic gives 0.0802, 0.0321, 0.0226 and 0.0217 (numerically verified, see Section 7).

5 A weak Potts field as a tilt

In this section the start is uniform again. Give each visit to state i the extra weight $e^{h_i/N}$, for a fixed vector $h \in \mathbb{R}^d$. Over N steps these weights multiply to $e^{h \cdot n/N}$, a factor of order 1, which is the scale of the constants a_k . So the field can reorder the faces. We study the *tilted occupation law*

$$\tilde{P}_{N,h}(n) = \frac{e^{h \cdot n/N} P_N(n)}{\sum_{m_1 + \dots + m_d = N} e^{h \cdot m/N} P_N(m)}. \quad (9)$$

We use the window coordinate $T = L - \frac{1}{2} \log L + t$. The normalizing sum cancels whenever we compare two bins. A modal phase is the support of the global modes, which need not be the face with most of the mass.

Proposition 5.1 (the field is a tilt). *Let $\Lambda_N(h) = \log \mathbb{E} \exp\langle h, K_N/N \rangle$.*

- (1) $\tilde{P}_{N,h}(n) = \exp(\langle h, n/N \rangle - \Lambda_N(h)) P_N(n)$. So $\tilde{P}_{N,h}$ is the exponential tilt of the law of the occupation rate K_N/N by the linear function $\langle h, \cdot \rangle$.
- (2) It is also the law of the counts when the law of the path is tilted by $\exp(\sum_{s=1}^N h_{X_s}/N)$. This is the Potts chain with free boundaries, coupling p/r per equal neighbor pair and field h/N at each site.
- (3) If $T \rightarrow \infty$, then $\Lambda_N(h) \rightarrow \bar{h} = \frac{1}{d} \sum_i h_i$.

Proof. Part (1) is (9) with its denominator written as $e^{\Lambda_N(h)}$. For (2), a word $x_1 \dots x_N$ has probability $\frac{1}{d} r^{N-1} (p/r)^{\sum_{s < N} \delta(x_s, x_{s+1})}$, and $\sum_s h_{x_s} = \langle h, n \rangle$ depends on the word only through its counts n . So tilting the words and then counting gives (1).

For (3), the chain is stationary with transition matrix $\sigma I + (1 - \sigma) \frac{1}{d} J$, whose m -th power is $\sigma^m I + (1 - \sigma^m) \frac{1}{d} J$. So $\mathbb{E} K_{N,a} = N/d$, and the indicators of state a at times s and $s + m$ have covariance $\sigma^m (\frac{1}{d} - \frac{1}{d^2})$. Summing over the two times gives, for $p < 1$,

$$\text{Var } K_{N,a} = \left(\frac{1}{d} - \frac{1}{d^2} \right) \left(N \frac{1 + \sigma}{1 - \sigma} - \frac{2\sigma(1 - \sigma^N)}{(1 - \sigma)^2} \right).$$

We bound this in two cases. Note that $1 - \sigma = dr$. If $1/d \leq p < 1$, then $0 \leq \sigma < 1$, the second term in the bracket is not positive, and $\text{Var}(K_{N,a}/N) \leq 2/(d^2 T)$. If $p < 1/d$, then $-1 \leq \sigma < 0$, the

bracket is at most $N + 4$, and $T \leq N$, so $\text{Var}(K_{N,a}/N) \leq 5/T$. In both cases K_N/N tends to the uniform vector in probability as $T \rightarrow \infty$. Since $\exp\langle h, K_N/N \rangle$ is bounded, bounded convergence gives the limit. $\square$

By (1), $|\log \tilde{P}_{N,h}(n) - \log P_N(n)| \leq 2 \max_i |h_i|$ for every n . This is of the order of the constants a_k , but invisible at first order, so a field of total size $o(\log N)$ does not change the exponents of Theorem 3.1(ii). For a support S with $|S| \geq 2$ put $\bar{h}_S = \frac{1}{|S|} \sum_{i \in S} h_i$.

Theorem 5.2 (weak-field modal phase diagram). *Put $T = L - \frac{1}{2} \log L + t$ and define the face scores*

$$\ell_S(t, h) = \begin{cases} h_i, & S = \{i\}, \\ (|S| - 1)(t - a_{|S|}) + \bar{h}_S, & |S| \geq 2. \end{cases}$$

The largest bin weight, taken relative to the untilted vertex and before the normalization in (9), satisfies, uniformly for (t, h) in compact sets,

$$\log \max_n \left\{ e^{h \cdot n/N} \frac{P_N(n)}{V_N} \right\} \longrightarrow \max_{\emptyset \neq S \subseteq \{1, \dots, d\}} \ell_S(t, h). \quad (10)$$

For fixed (t, h) and all large N , every global mode has a support that maximizes the limiting score. If the maximizing support is unique, all global modes have exactly that support. On a selected support S with $|S| \geq 2$, their locations satisfy

$$\max_{i \in S} \left| \frac{n_i}{N} - \frac{1}{|S|} \right| = o((\log N)^{-1/2}).$$

This does not identify the maximizing vectors exactly, and equal limiting scores do not imply that both supports hold modes at finite N .

If the fields are sorted as $h_1 \geq \dots \geq h_d$, the maximum of the scores is the upper envelope of the d lines

$$\ell_1(t) = h_1, \quad \ell_k(t) = (k - 1)(t - a_k) + \frac{1}{k} \sum_{i=1}^k h_i, \quad 2 \leq k \leq d. \quad (11)$$

Away from ties, the winning support size is nondecreasing in t . If the fields are strictly ordered, the winning supports are the nested sets $\{1, \dots, k\}$. Every support size $k \in \{2, \dots, d - 1\}$ is the unique winning size on an open interval of t for suitable fixed fields. So proper-face phases persist under a field of order $1/N$ per step.

So the field shifts the line of each face by the field averaged over the face. The reason is that on a support S the ratio of a bin to a vertex has a Gaussian profile of width $1/\sqrt{L}$ around the center of the face (Lemma B.1), where the tilt is $e^{\bar{h}_S + o(1)}$. The proof is in Appendix B.

Example 5.3 (a persistent triangular edge phase). For $d = 3$ and $h = (0, 0, -H)$ the three competing scores are 0, $t - a_2$ and $2(t - a_3) - H/3$. If $H > 6(a_2 - a_3)$, the edge $\{1, 2\}$ wins strictly for

$$a_2 < t < H/3 + 2a_3 - a_2,$$

and for every fixed t in this interval all global modes of the tilted law lie on that edge for large N . Without a field no proper face holds a mode in the window (Corollary 3.9).

One field can make every support size win in turn, from one state to all d .

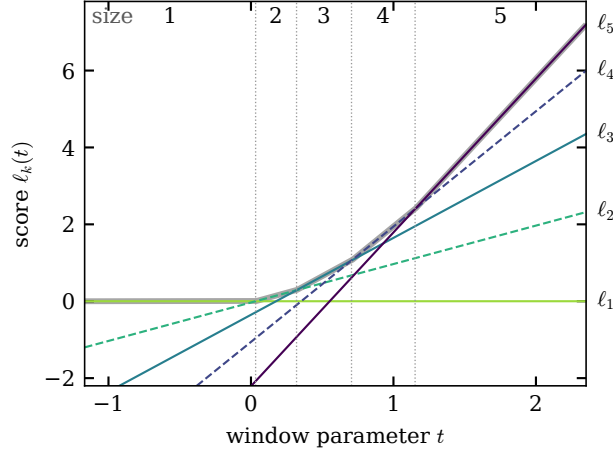


Figure 2: The lines (11) for $d = 5$ and $h_i = -(i-1)^2$. The gray band is their upper envelope. The dotted lines are $t_1 < \dots < t_4$, and the number at the top of each interval is the size k of the winning support $\{1, \dots, k\}$.

Corollary 5.4 (one field realizes the full face hierarchy). *Fix $d \geq 2$ and $H > 3/4$, and let $h_i = -H(i-1)^2$ for $1 \leq i \leq d$. In particular $h_i = -(i-1)^2$ works in every fixed dimension. Put*

$$c_k = (k + \tfrac{1}{2}) \log k - \frac{k-1}{2} \log(4\pi), \quad t_k = c_k - c_{k+1} + \frac{H(4k-1)}{6} \quad (1 \leq k < d),$$

and $t_0 = -\infty$, $t_d = +\infty$. Then $t_1 < \dots < t_{d-1}$, and for every $k \in \{1, \dots, d\}$ the support $\{1, \dots, k\}$ is the unique winning support on the nonempty interval $t_{k-1} < t < t_k$. So for each fixed t in this interval and all large N , every global mode of the tilted law has exactly those k positive coordinates.

Here $c_1 = 0$ and $c_k = \log D_k$ for $k \geq 2$. Figure 2 shows the lines for $d = 5$, and the proof is in Appendix B.

6 Near the boundary of the simplex

So far every positive coordinate was of order N (Theorem 3.3 needs each to be at least εN). In this section we no longer assume this, and first give an upper bound that holds for every positive count vector (Theorem 6.2). Then we find the scale at which a small coordinate starts to matter. It is N/L^2 when at least two coordinates are of order N (Remark 6.3). Near a vertex the crossing moves to odds of order $N^{-1/2}$, and the small coordinates alone decide it (Subsection 6.1).

The start is uniform, but $S_n(u)$ does not involve the start, and Proposition 2.2 gives $P_N^\mu(n) \leq (\max_i \mu_i) p^{N-1} S_n(u)$. So every upper bound below also holds for a general start.

Lemma 6.1 (a uniform Laguerre–Bessel inequality). *For every integer $a \geq 1$ and real $w > 0$,*

$$0 \leq 1 - \frac{B_a(w)}{w\Phi(aw)} \leq \frac{w}{2}, \quad B_a(w) = \sum_{m=1}^a \binom{a-1}{m-1} \frac{w^m}{m!}. \quad (12)$$

By [15, Eq. 18.5.12], $B_a(w) = (w/a)L_{a-1}^{(1)}(-w)$ with the Laguerre polynomial $L_{a-1}^{(1)}$, so the lemma is a quantitative form of the limit of Laguerre polynomials to Bessel functions. The proof, in Appendix C, compares coefficients with Lemma A.2.

Theorem 6.2 (a uniform Bessel majorant on every simplex stratum). *Let $n_1, \dots, n_k$ be arbitrary positive integers with sum N , and let $0 < u < 1$, $v = u/(1 - u)$. Define*

$$\mathcal{Q}_n(u) = (1 - u)^{N-1} v^{k-1} \int_0^\infty e^{-t} t^k \prod_{i=1}^k \Phi(n_i v t) dt.$$

Then, with no lower bound on any n_i/N ,

$$0 \leq 1 - \frac{S_n(u)}{\mathcal{Q}_n(u)} \leq k(k+1)v + \frac{k}{2}v^2 \left(\sum_i \sqrt{n_i} \right)^2 \leq k(k+1)v + \frac{k^2}{2}Nv^2. \quad (13)$$

So this approximation has relative error $O_k(L^2/N)$ whenever $u = O(L/N)$, uniformly over every positive composition of N .

For the central bin of the segment, $\mathcal{Q}_n$ is a bound of the same kind as the uniform Bessel bound of Paper A [2, Theorem 3.3]. The proof is in Appendix C. It applies Lemma 6.1 to each factor of an integral form of S_n , and an integration by parts bounds the resulting mean.

Remark 6.3 (the scale $N/(\log N)^2$). A small coordinate a adds its own runs. Each run needs about two switches, which gives a factor u^2 , and it can go in about N places. So what matters is aNu^2 . At the crossings of Section 3 the value Nu is of order L , and then aNu^2 is of order aL^2/N . So when at least two coordinates are of order N , a rare count starts to matter once it has size $\rho N/L^2$ with ρ fixed.

Appendix C proves this. Take $k_1 \geq 2$ large coordinates with fractions β_i among themselves and ℓ rare counts $a_j = \rho_j N/L^2$. With $z = Nu$, Theorem C.1 gives $S_n(z/N)$ up to a factor $1 + O(L^{-2} + L^2/N)$. Each rare count contributes a factor $\Phi(A^2 a_j z^2/N)$, where $A = \sum_i \sqrt{\beta_i}$, and the complete correction of relative order $1/L$ is a factor $e^{\mathcal{E}(z)}$ with three terms. The theorem also gives the crossing as the root of an implicit equation, and as an explicit expansion in L , which converges slowly. Rare counts with $a_j = o(N/L^2)$ change that expansion only by $o(1)$ (Corollary C.3). With only one large coordinate the growth factor $e^{(A^2-1)z}$ is gone, since $A = 1$. Then the crossing moves to u of order $b^{-1/2}$, where b is the large count. This is the case of Subsection 6.1.

6.1 Near a vertex

Now let one count b be large and let the other positive counts $a_1, \dots, a_\ell$ be small compared with b , with $\ell \geq 1$ fixed. In this subsection and in Appendix D we put $|a| = a_1 + \dots + a_\ell$. These rare counts are not the constants a_k of Section 3, which do not appear here. With B_a as in Lemma 6.1, put

$$\mathcal{H}(y) = \prod_{i=1}^\ell B_{a_i}(y), \quad \kappa(y) = \frac{y\mathcal{H}'(y)}{\mathcal{H}(y)}.$$

The polynomial $\mathcal{H}$ has positive coefficients, $\mathcal{H}(0) = 0$ and $\mathcal{H}(1) \geq 1$, so $\mathcal{H}(y_a) = 1$ has exactly one positive root y_a , and $y_a \leq 1$. Write $S_{(b,a)}$ for S_n with $n = (b, a_1, \dots, a_\ell)$. It is a polynomial in u with nonnegative coefficients, zero constant term and a positive coefficient at u^ℓ , so $S_{(b,a)}(u) = 1$ has exactly one positive root $u_{b,a}$.

Each rare run again needs about two switches and has about b places, so the rare counts enter through bu^2 . For fixed rare counts the bounds of Lemma D.2 give $S_{(b,a)}(\sqrt{y/b}) \rightarrow \mathcal{H}(y)$ as $b \rightarrow \infty$. So the crossing is near $\sqrt{y_a/b}$, and Nu is of order $\sqrt{N}$ instead of L . The next theorem shows that this stays correct uniformly as long as $|a| = o(b)$.

Theorem 6.4 (all growing rare counts near a vertex). *Fix $\ell \geq 1$ and put*

$$c_\ell = \ell + \sqrt{\ell} + 1, \quad \varepsilon = \sqrt{\frac{|a| + 1}{b}}, \quad \delta = 16c_\ell \varepsilon.$$

For all positive counts $b, a_1, \dots, a_\ell$ with $\delta \leq 1/2$,

$$(1 - \delta)\sqrt{y_a/b} \leq u_{b,a} \leq (1 + \delta)\sqrt{y_a/b}, \quad \mathcal{H}(y_a) = 1. \quad (14)$$

In particular

$$\left| \frac{u_{b,a}}{\sqrt{y_a/b}} - 1 \right| \leq 16c_\ell \sqrt{\frac{|a| + 1}{b}} \quad \text{whenever} \quad \sqrt{\frac{|a| + 1}{b}} \leq \frac{1}{32c_\ell}, \quad (15)$$

so $u_{b,a} = \sqrt{y_a/b}(1 + o(1))$ as $(|a| + 1)/b \rightarrow 0$. The constants are not optimized.

The proof is in Appendix D. It rests on two facts. First, $\log \mathcal{H}$ is concave, and for $0 < y \leq 1$ its elasticity $\kappa(y)$ lies between $\max\{\ell, \frac{1}{2}\sqrt{|a|y}\}$ and $\ell + \sqrt{\ell|a|y}$ (Lemma D.1). Second, $S_{(b,a)}(u) = (1 - u)^{|a|}\mathbb{E}[\mathcal{H}(v\mathcal{X})]$ exactly, where $v = u/(1 - u)$ and $\mathcal{X}$ is a gamma variable plus a compound binomial sum with mean $2 + (b - 1)u$ (Lemma D.2). Jensen's inequality then gives $S_{(b,a)}(u) \geq (1 - u)^{|a|}\mathcal{H}(bu^2)$, which exceeds 1 at $u = (1 + \delta)\sqrt{y_a/b}$. The tangent line of $\log \mathcal{H}$ at bu^2 and the moment generating function of $\mathcal{X}$ give an upper bound that is below 1 at $u = (1 - \delta)\sqrt{y_a/b}$.

7 Verification and open questions

Data and code availability. No proof in this paper relies on computation. The code and the Lean files are in the folder `kagey-problems/problem131` of <https://github.com/apemm/Kagey-Problems> (release `paperB-arxiv-v1`). The checks for this paper are described in `SUPPLEMENT.md`. The scripts are in the subfolder `paperB` and use only the Python standard library. The command `python -B verify_all.py --paper b` in `kagey-problems/problem131` runs all of them except `verify_Btie_repair.py`, which also runs when `--full` is added. The scripts compare the exact law with dynamic programs and with enumeration of all words, and check the crossings and the constants. The statements marked as numerically verified are checked as follows. For the error quoted after Proposition 4.3, `verify_general_start.py` computes the start weights from the run form and from the refresh form in double precision. For the finite- N phases after Example 4.9, dynamic programs over all bins run at $N = 90$ in `verify_general_start.py` and at $N = 300$ and 1002 in `verify_Btie_repair.py`. For Remark 4.10 the two ends of the window are found by bisection, at $N = 300$ in `verify_Btie_repair.py` and at $N \geq 3000$ in `verify_general_start.py`. The scripts `verify_frontier_modes.py`, `verify_frontier_boundary.py` and `verify_vertex_completion.py` check the field and the identities near the boundary against word enumeration. The numbers in Remark C.2 come from `verify_crossover_factor.py`. For $N \leq 2 \cdot 10^6$ it computes $S_{(n,a)}$ in three ways, by the run form and by a quadrature of (21) in double precision and by the sum (3) in 60-digit decimal arithmetic, and for larger N by the last two. They agree to about 10^{-14} in $\log S$. None of these checks is interval-certified.

Many of the exact identities and finite bounds were also formalized in Lean 4, in `kagey-problems/problem131/lean/PaperB`, where `MAP.md` matches each statement with its Lean theorem. These include the exact law for the uniform start, balancing, the ordering of the constants a_k , most of the arithmetic of Corollary 5.4, and the bracket near a vertex. The asymptotic results and the results for a nonuniform start were not formalized.

Open questions. Several questions remain open. The discrete correction term $-\frac{(k-1)^2}{2}z^2/N$ and the second-order ends of the window in Remark 4.10 are not proved. Neither is the failure of balancing at every large N for a start that is not constant on the face. Corollary 3.9 has no effective N_0 , and by Example 3.10 it does not extend to every N . Nothing is said here about the modes when T is much larger than L . Near a vertex the leading crossing is known uniformly for rare counts of total $o(b)$, but neither the next term uniformly in that range nor the case of a nonuniform start is known. The crossover with a start that is not constant on the support, and a number of states that grows with N , are also left open. Switching that is not uniform is treated in a paper of the author that is in preparation.

Acknowledgments. I thank Peter Kagey for posing the problem, and Chase Hameetman and Aidan Deshong for insightful conversations about some of the proofs in this paper.

A note on the use of AI

I wrote the proofs and derivations of Sections 2 and 3 by hand, without AI assistance, before any revision. I then used Claude Opus 5.5 (Anthropic, through Claude Code) in September and October 2026 to check everything I had written for typos and for statements that were technically wrong. It found a number of gaps, which I fixed, along with occasional wrong references and unclear wording. I decided every revision and went over each one by hand. The model also drafted rewordings of many passages, which I reviewed and edited. For the later sections I brainstormed conjectures with the model, and several results came from it together with proofs, namely the tree form of the constants in Proposition 3.6, the start-law results (Lemma 4.2, Proposition 4.3, Theorem 4.6, Corollary 4.7 and Example 4.9) and the full correction factor in Theorem C.1. I rechecked and rederived these proofs by hand. I also checked the proofs with GPT-6 Astra (OpenAI), which found nothing beyond the earlier check.

The Lean 4 formalization in `kagey-problems/problem131/lean/PaperB` was generated by the model, then revised and checked by hand against the corresponding results (see `MAP.md` in that folder). Parts of the Python verification scripts were also written by the model, which cross-checked them against each other. Lastly, I used it for literature search, namely to find and verify sources beyond the ones I had first checked on my own.

All mathematical claims, proofs and computations in this paper are my responsibility.

A Proofs for Sections 2 and 3

A.1 Facts about Bessel series

The weights $x^m/(m!(m+1)!)$ of the series Φ define a probability law on the integers $m \geq 0$. Its moments control every Laplace expansion in this paper.

Lemma A.1 (the Bessel law). *For $x \geq 0$ let J be a random variable with $\mathbb{P}(J = m) = x^m/[m!(m+1)!\Phi(x)]$ for $m \geq 0$ (so $J = 0$ when $x = 0$), and put $\psi(x) = \mathbb{E}J$. Then*

$$(i) \quad \mathbb{E}[J(J+1)] = x,$$

$$(ii) \quad \mathbb{E}[J(J-1)J(J+1)] = x^2,$$

$$(iii) \quad \psi(x) = x\Phi'(x)/\Phi(x), \text{ and } \psi(x) = \sqrt{x} I_2(2\sqrt{x})/I_1(2\sqrt{x}) \text{ for } x > 0,$$

(iv) $x\psi'(x) = \text{Var } J$, so $0 \leq \psi'(x) \leq 1$ for $x > 0$, and $\psi'(0) = \frac{1}{2}$,

(v) $0 \leq \psi(x) \leq \min(x/2, \sqrt{x})$,

(vi) $0 \leq (\log \Phi)'(x) = \psi(x)/x \leq \frac{1}{2}$ for $x > 0$.

Proof. (i) We have $\sum_{m \geq 1} m(m+1)x^m/[m!(m+1)!] = \sum_{m \geq 1} x^m/[(m-1)!m!] = x\Phi(x)$. (ii) For $m \geq 2$, $m(m-1)/m! = 1/(m-2)!$ and $m(m+1)/(m+1)! = 1/(m-1)!$, so the sum is $\sum_{m \geq 2} x^m/[(m-2)!(m-1)!] = x^2\Phi(x)$. (iii) The first form is the definition of the mean. Termwise, $\Phi'(x) = \sum_{i \geq 0} x^i/[i!(i+2)!] = I_2(2\sqrt{x})/x$ by the series of I_2 [15, Eq. 10.25.2], which gives the second. (iv) Since $\log \mathbb{P}(J = m) = m \log x - \log \Phi(x)$ plus a term free of x , we have $x \frac{d}{dx} \log \mathbb{P}(J = m) = m - \psi(x)$. So $x\psi'(x) = \mathbb{E}[J(J-\psi)] = \text{Var } J$, and $\text{Var } J \leq \mathbb{E}J^2 \leq \mathbb{E}[J(J+1)] = x$. From the series, $\psi(x) = x(\frac{1}{2} + \frac{x}{6} + \dots)/(1 + \frac{x}{2} + \dots) = \frac{x}{2} + O(x^2)$, so $\psi'(0) = \frac{1}{2}$. (v) For integers $J \geq 0$ we have $J \leq J(J+1)/2$, so $\psi \leq x/2$ by (i). Also $\psi^2 \leq \mathbb{E}J^2 \leq x$. (vi) follows from (iii) and (v). $\square$

Lemma A.2 (clipped product deficit). *For any finite list of nonnegative real numbers $t_1, \dots, t_m$, with $(x)_+ = \max(x, 0)$,*

$$0 \leq 1 - \prod_{j=1}^m (1 - t_j)_+ \leq \sum_{j=1}^m t_j.$$

Proof. If some $t_j \geq 1$, the product is zero and the sum is at least one. Otherwise all factors lie in $[0, 1]$, and induction on m follows from $1 - (1-a)(1-b) = a + b - ab \leq a + b$ for $a, b \in [0, 1]$. The empty product gives equality. $\square$

Lemma A.3 (Gaussian averages of the factors Φ). *Fix $q \geq 0$, an integer $\ell \geq 0$ and $X \geq 0$. Let $w_0 \geq 1$ and $c_1, \dots, c_\ell \geq 0$ with $\xi_j = c_j w_0^2 \leq X$, and put $F(w) = w^q \prod_{j=1}^\ell \Phi(c_j w^2)$. Let $J_1, \dots, J_\ell$ be independent, with J_j distributed as J in Lemma A.1 with $x = \xi_j$. Put $|J| = J_1 + \dots + J_\ell$ and $\mu_q = \mathbb{E}[(q+2|J|)(q+2|J|-1)]$. Then*

$$\frac{1}{\sqrt{\pi}} \int_{-w_0/2}^{w_0/2} e^{-y^2} F(w_0 + y) dy = F(w_0) \left(1 + \frac{\mu_q}{4w_0^2} + O(w_0^{-4}) \right),$$

and $F(w_0 + y) = O(F(w_0))$ for $|y| \leq w_0/2$. The constants depend only on q, ℓ and X . Moreover

$$\mu_q = q^2 - q + 4 \sum_j \xi_j + (4q - 6) \sum_j \psi(\xi_j) + 8 \sum_{i < j} \psi(\xi_i) \psi(\xi_j).$$

Proof. Step 1. Derivatives as moments. Expanding each Φ gives $F(w) = \sum_m \kappa_m w^{q+2|m|}$, where $m = (m_1, \dots, m_\ell)$ runs over vectors of nonnegative integers, $|m| = \sum_j m_j$ and $\kappa_m = \prod_j c_j^{m_j} / [m_j!(m_j+1)!] \geq 0$. The weights $\kappa_m w_0^{q+2|m|} / F(w_0)$ are the joint law of $(J_1, \dots, J_\ell)$. Hence $F^{(i)}(w_0) = F(w_0) \mathbb{E}[(q+2|J|)_i] w_0^{-i}$, where $(y)_i = y(y-1)\dots(y-i+1)$. In particular $F''(w_0) = \mu_q w_0^{-2} F(w_0)$.

Step 2. The fourth derivative. Let $w = w_0 + y$ with $|y| \leq w_0/2$. For every $\eta \geq 0$ we have $w^{\eta-4} \leq 16(3/2)^\eta w_0^{\eta-4}$ (consider $\eta \geq 4$ and $\eta < 4$ separately) and $|(\eta)_4| \leq \eta^4 + 1$. Applying this with $\eta = q + 2|m|$ termwise gives

$$|F''''(w)| \leq 16 \left(\frac{3}{2} \right)^q w_0^{q-4} \prod_j \Phi\left(\frac{9\xi_j}{4} \right) \tilde{\mathbb{E}}[(q+2|\tilde{J}|)^4 + 1],$$

where the $\tilde{J}_j$ are independent with the parameters $9\xi_j/4$ in place of ξ_j . By Lemma A.1(i) and (ii), $\mathbb{E}J^4 = x^2 + \mathbb{E}J^2 \leq x^2 + x$. Hence $\tilde{\mathbb{E}}[(q+2|\tilde{J}|)^4]$ is bounded in terms of q , ℓ and X . Since $\Phi \geq 1$ we have $F(w_0) \geq w_0^q$, so $F''''(w) = O(w_0^{-4}F(w_0))$. The same argument without derivatives bounds $F(w_0 + y)$.

Step 3. Taylor expansion. Expand $F(w_0 + y)$ to third order in y with the Lagrange remainder. The odd terms integrate to zero over the symmetric interval. The integrals of e^{-y^2} and $y^2 e^{-y^2}$ over $|y| \leq w_0/2$ differ from $\sqrt{\pi}$ and $\sqrt{\pi}/2$ by $O(w_0 e^{-w_0^2/4})$, the number μ_q is bounded, and the remainder term is $O(w_0^{-4}F(w_0))$. This proves the expansion.

Step 4. The formula for μ_q . Put $\psi_j = \psi(\xi_j) = \mathbb{E}J_j$. Then $\mathbb{E}J_j^2 = \xi_j - \psi_j$ by Lemma A.1(i), and independence gives $\mathbb{E}[(q+2|J|)^2] = q^2 + 4q \sum_j \psi_j + 4 \sum_j (\xi_j - \psi_j) + 8 \sum_{i < j} \psi_i \psi_j$. Subtracting $\mathbb{E}[q+2|J|] = q + 2 \sum_j \psi_j$ proves it. $\square$

A.2 Proofs for Section 3

Proof of the exact assertions in Theorem 3.2. Let n be a balanced vector with k positive coordinates, so that $C_{N,k}/V_N = S_n(u)$ with $u = (1-p)/((d-1)p)$.

Step 1. One crossing. To count $W_n(j)$, choose a sequence of $j+1$ run colors in which successive colors differ. If color i appears m_i times, its run lengths can be chosen in $\binom{n_i-1}{m_i-1}$ ways. So every coefficient of S_n is nonnegative, and the first nonzero one is $W_n(k-1) = k!$, from the $k!$ orders of k single runs. Hence S_n increases strictly from 0 to ∞ on $u > 0$ and equals 1 exactly once. As p increases over $(0, 1)$, u decreases over $(0, \infty)$, which gives the unique $p_{N,k}$ and the two inequalities.

Step 2. Monotonicity in N . Choose the balanced vectors for increasing N by adding 1 to one of the smallest coordinates. Each coefficient $W_n(j)$ is then nondecreasing. To see this, map a word to the word with an extra copy of the incremented color just before its first occurrence. This keeps the number of changes, and it is injective, since deleting the first letter of the first run of that color inverts it. The coefficient of u^k is

$$W_n(k) = \frac{k!(k-1)}{2} \sum_i (n_i - 1) = \frac{k!(k-1)}{2} (N - k).$$

To see this, note that exactly one run color appears twice. For each choice of it there are $(k+1)!/2 - k! = k!(k-1)/2$ sequences of run colors in which successive colors differ (all orders, minus those with the two copies adjacent), and $n_i - 1$ ways to split its n_i visits into two runs. This coefficient increases strictly with N . So at every $u > 0$ the value $S_n(u)$ increases strictly with N , the root of $S_n(u) = 1$ decreases strictly with N , and $p = 1/(1 + (d-1)u)$ increases.

Step 3. The first crossing. If $N = k$, every coordinate is 1 and $S_n(u) = k!u^{k-1}$, which gives $p_{k,k}$. Since the root decreases with N , every root has $u \leq (k!)^{-1/(k-1)} < 1$, and $u < 1$ means $r < p$, that is, $p > 1/d$. $\square$

The asymptotic assertions rest on a Laplace expansion of the positive sum (3). It is the case with no rare counts of the crossover theorem in Appendix C.

Lemma A.4 (no rare counts). *Fix $k \geq 2$, $\varepsilon > 0$ and $0 < c < C$. Let n be a positive count vector with k coordinates and total N , put $\beta = n/N$, and assume that every $\beta_i \geq \varepsilon$. Put $A = \sum_i \sqrt{\beta_i}$, $g = A^2 - 1$ and $s = k - 1$, let $D(\beta)$ be as in Theorem 3.3, and put*

$$\gamma_0(\beta) = \frac{k(k+2)}{16A^2} - \frac{3}{16A} \sum_{i=1}^k \beta_i^{-1/2}.$$

Then, uniformly for these n and $cL \leq z \leq CL$,

$$S_n(z/N) = D(\beta) \frac{z^{s/2} e^{gz}}{N^s} e^{\gamma_0(\beta)/z} \left(1 + O\left(\frac{1}{L^2} + \frac{L^2}{N}\right)\right). \quad (16)$$

Proof. This is Theorem C.1 with $\ell = 0$ and $k_1 = k$. Then $M = N$, $\eta = s/2$, the constant $D_{k_1}(\beta)$ is $D(\beta)$, there are no factors Φ , and $\mathcal{E}(z) = \gamma_0/z$ with γ_0 as above, since $(2\eta + 1)(2\eta + 3) = k(k + 2)$. The proof of that theorem uses only Theorem 6.2, the expansion of I_1 and Lemma A.3, and nothing from Section 3. $\square$

Proof of the asymptotic assertions in Theorems 3.1 and 3.2. Put $s = k - 1$ and let n be balanced, so that $\beta_i = n_i/N = 1/k + O(1/N)$.

Step 1. The balanced ratio (6). At the center $A^2 - 1 = s$, $D(\beta) = D_k$ and $\gamma_0 = (k + 2)/16 - 3k/16 = -s/8$. On the plane $\sum_i \beta_i = 1$ the function A^2 has a critical point at the center, so $A^2 - 1 = s + O(N^{-2})$, while $D(\beta)$ and $\gamma_0(\beta)$ change by $O(1/N)$. Since $z = O(L)$, Lemma A.4 with $\varepsilon = 1/(2k)$ and $e^{-s/(8z)} = 1 - s/(8z) + O(z^{-2})$ gives (6).

Step 2. Parts (i) to (iii). Let $cL \leq T \leq CL$ and $u = r/p$, so that $z = T/p$ lies in $[cL, 2CL]$ for large N and $z - T = (d - 1)Tz/N = O(L^2/N)$. Take logarithms in (6), with c and $2C$ in place of c and C , and replace z by T . This changes each term by $O(T^2/N)$, and $\log D_k = -sa_k$, so we get (i). For (ii), $\log V_N = (N - 1) \log(1 - (d - 1)T/N) - \log d = -(d - 1)T - \log d + O(T^2/N)$. With $T = \tau L$ and (i) we get $\log C_{N,k} = -(d - 1)\tau L + (k - 1)(\tau - 1)L + O(\log L)$, which is (ii), and the case $k = 1$ is the formula for $\log V_N$. For (iii), $T = L - \frac{1}{2} \log L + t$ gives $\Psi_N(T) = t + \frac{1}{2} \log(T/L) = t + O(\log L/L)$ uniformly on compact t sets, and the terms $1/(8T)$ and the error tend to 0.

Step 3. The crossing. Part (i) at $T = L/2$ and at $T = 2L$ shows that the ratio tends to 0 and to ∞ . Since $S_n(u)$ increases with u and u increases with T , the root lies in $[L/2, 2L]$ for large N , and (i) applies there with $c = \frac{1}{2}$ and $C = 2$. The same argument at $T = (1 \pm \varepsilon)L$ gives $T_{N,k}/L \rightarrow 1$. At the root the left side of (i) is 0, and dividing by s gives the implicit equation, since $T^2/N = o(L^{-2})$. For the explicit form put $z_* = L - \frac{1}{2} \log L + a_k + e_L/L$ with $e_L = \frac{1}{4} \log L - \frac{1}{2} a_k + \frac{1}{8}$. Since $\log z_* = \log L - \frac{\log L}{2L} + \frac{a_k}{L} + O(\frac{(\log L)^2}{L^2})$, we get $z_* + \frac{1}{2} \log z_* - 1/(8z_*) = L + a_k + O((\log L)^2/L^2)$. The derivative of $z + \frac{1}{2} \log z - 1/(8z)$ is at least 1 for $z > 0$, so $T_{N,k} - z_* = O((\log L)^2/L^2)$, which is (5).

Step 4. Part (iv). We have $a_k = \frac{1}{2} \log(4\pi) - f(k)$ with $f(x) = (x + \frac{1}{2}) \log x / (x - 1)$. For $x \geq 2$ the sign of $f'(x)$ is the sign of $\Delta(x) = 2x^2 - x - 1 - 3x \log x$. Here $\Delta(2) = 5 - 6 \log 2 > 0$, and $\Delta'(x) = 4x - 4 - 3 \log x > 0$ for $x \geq 2$, since $\Delta'(2) > 0$ and $\Delta''(x) = 4 - 3/x > 0$. So f increases strictly and a_k decreases strictly. $\square$

Proof of Theorem 3.3. The polynomial S_n has nonnegative coefficients and lowest term $k!u^{k-1}$, so $S_n(u) = 1$ has a unique positive root. Lemma A.4 with $\beta = \alpha$ gives (7), since $\gamma_0(\alpha)$ is bounded when every $\alpha_i \geq \varepsilon$ and $e^{\gamma_0/z} = 1 + O(1/z)$.

For the root, note that $A^2 = 1 + 2 \sum_{i < j} \sqrt{\alpha_i \alpha_j}$, so $g \in [2\varepsilon, s]$, and $D(\alpha)$ has positive lower and finite upper bounds depending on k and ε . Put $c_0 = \frac{1}{2}$ and $C_0 = s/\varepsilon$. By (7), $\log S_n(c_0 L/N) \leq (gc_0 - s)L + O(\log L) \leq -\frac{s}{2}L + O(\log L) < 0$ and $\log S_n(C_0 L/N) \geq (gC_0 - s)L + O(\log L) \geq sL + O(\log L) > 0$ for large N . So the root has $Nu_n \in [c_0 L, C_0 L]$, and taking logarithms in (7) there gives

$$gz + \frac{s}{2} \log z = sL - \log D(\alpha) + O_{k,\varepsilon}(z^{-1} + z^2/N), \quad z = Nu_n.$$

First $z = (s/g)L + O_{k,\varepsilon}(\log L)$. Then $\frac{s}{2} \log z = \frac{s}{2} \log L + \frac{s}{2} \log(s/g) + O(\log L/L)$, and solving for z gives (8). The derivative of the left side is at least $g \geq 2\varepsilon$, so the error is preserved on inversion.

Finally, the functions of α in the formula have bounded derivatives on the set $\{\alpha_i \geq \varepsilon/2\}$. So moving α by $O(1/N)$ changes the expression by $O(L/N)$, which the error term absorbs. $\square$

Proof of Lemma 3.4. Note that $(\mathcal{L}x)_i = \sum_j c_{ij}(x_i - x_j)$ and $\mathcal{L}\mathbf{1} = 0$. Take θ with $\theta_1 = 0$ and put $\zeta = \text{diag}(\varpi)f$. Then $\sum_i \zeta_i = 0$, $\mathcal{L}\chi = \zeta$ and the form is $2\zeta^\top \chi$. The vector $\tilde{\chi} = \chi - \chi_1 \mathbf{1}$ also solves $\mathcal{L}\tilde{\chi} = \zeta$, with $\zeta^\top \tilde{\chi} = \zeta^\top \chi$ and $\tilde{\chi}_1 = 0$. So rows 2 to k give $\mathcal{L}_{(1)}\tilde{\chi}' = \zeta'$, where primes delete the first coordinate, and the form is $2\zeta'^\top \mathcal{L}_{(1)}^{-1}\zeta'$. As $\theta_1 = 0$, we have $\zeta' = (D' - \varpi'\varpi'^\top)\theta'$ with $D' = \text{diag}(\varpi_2, \dots, \varpi_k)$. Hence $\Sigma = 2(D' - \varpi'\varpi'^\top)\mathcal{L}_{(1)}^{-1}(D' - \varpi'\varpi'^\top)$. The matrix determinant lemma gives $\det(D' - \varpi'\varpi'^\top) = \prod_{i \geq 2} \varpi_i (1 - \sum_{i \geq 2} \varpi_i) = \prod_i \varpi_i$, and the matrix-tree theorem gives $\det \mathcal{L}_{(1)} = \text{tree}(c) > 0$. For nonnegative integer c_{ij} this is the matrix-tree theorem for multigraphs [14]. Both sides are polynomials in the c_{ij} , so they agree for all real c_{ij} . $\square$

Proof of Proposition 3.6. (1) For $i \neq j$ we have $\alpha_i G_{ij}^\alpha = \phi_i \phi_j$, which is symmetric, so α is reversible and hence stationary. Also $(Q_k \phi)_i = A - k\phi_i$, so $\theta_{\alpha,i} = k - A/\phi_i$, and $Q_k + \text{diag}(\theta_\alpha)$ has off-diagonal entries 1 and diagonal entries $1 - A/\phi_i$. It is symmetric and kills $\phi > 0$, so 0 is its Perron root. Conjugating by $\text{diag}(\phi)$ gives the rates ϕ_j/ϕ_i . If G^α were a Doob transform of Q_k , then ϕ would be an eigenvector of Q_k , so $A - k\phi_i = \lambda\phi_i$ for all i and α would be uniform. (2) is Lemma 3.5 with $x_i = \phi_i$. (3) By (2), $\sqrt{\text{tree}(\alpha)}/\prod_i \alpha_i = (\prod_i \alpha_i)^{-3/4} A^{k/2-1}$, which gives the first form. Lemma 3.4 with $\varpi = \alpha$ gives $\det(2\pi\Sigma_\alpha) = (4\pi)^{k-1} (\prod_i \alpha_i)^2 / \text{tree}(\alpha)$, which gives the second. (4) At the center $\mathcal{L}_{(1)} = I - J/k$ and $D' - \alpha'\alpha'^\top = (I - J/k)/k$ in the formula $\Sigma = 2(D' - \alpha'\alpha'^\top)\mathcal{L}_{(1)}^{-1}(D' - \alpha'\alpha'^\top)$ from the proof of Lemma 3.4. This gives Σ_k . Substituting $\text{tree}(\alpha) = 1/k$ and $\prod_i \alpha_i = k^{-k}$ into (3) gives D_k . (5) is the same computation with k_1 and β . $\square$

Proof of Proposition 3.7(2) and (3). Let $u = z/N$ with $cL \leq z \leq CL$, and put $s = k - 1$.

Step 1. The mass. In (1) the term $j = k$ is $k(1 + (k-1)z/N)^{N-1} = k e^{(k-1)z + O(z^2/N)}$, and each term with $j < k$ is smaller by a factor $O(e^{-z})$. This gives the first expression.

Step 2. The Hessian. Write $\alpha' = (\alpha_2, \dots, \alpha_k)$ and $g(\alpha) = A^2 - 1$ as in Theorem 3.3. In full coordinates the Hessian of A^2 at the center is $(k/2)J - (k^2/2)I$. Pulling it back along $w = (-\sum_i v_i, v)$ gives $-\nabla^2 g = (k^2/2)(I + J)$ in the coordinates α' , and its inverse is $(2/k^2)(I - J/k) = \Sigma_k$.

Step 3. The bins near the center. Fix a small $\eta > 0$ and let K_η be the set of α with $|\alpha_i - 1/k| \leq \eta$ for all i . On K_η , Theorem 3.3 gives $S_n = D(\alpha)z^{s/2}e^{g(\alpha)z}N^{-s}(1 + O(z^{-1} + z^2/N))$ uniformly, with $\alpha = n/N$. The points α' form a lattice with cells of volume N^{-s} , and on each cell g changes by $O(1/N)$ and D by a factor $1 + O(1/N)$. So the sum over $n/N \in K_\eta$ is N^s times the integral of the same expression over K_η , times $1 + O(z/N)$, up to the cells near the boundary of K_η . There $g \leq k - 1 - c_\eta$ for some $c_\eta > 0$, so these cells are negligible. The function g has its strict maximum $k - 1$ at the center, and D is continuous there with $D = D_k$. So Laplace's method gives

$$\int_{K_\eta} D(\alpha) e^{g(\alpha)z} d\alpha' = D_k e^{(k-1)z} \left(\frac{2\pi}{z}\right)^{s/2} (\det \Sigma_k)^{1/2} (1 + o(1)).$$

Multiplied by $z^{s/2}$ this is the second expression, provided the bins off K_η are negligible.

Step 4. The other bins. For them, Theorem 6.2 and $\Phi(x) \leq e^{2\sqrt{x}}$ give, with $Y = Nv$ and $A_n = \sum_i \sqrt{n_i/N}$,

$$S_n(u) \leq (1-u)^{N-1} v^s \int_0^\infty e^{-t+2A_n\sqrt{Y}t} t^k dt \leq C' \frac{(1+z)^{2k}}{N^s} e^{(A_n^2-1)z + O(z^2/N)},$$

where we wrote $-t + 2A_n\sqrt{Y}t = A_n^2Y - (\sqrt{t} - A_n\sqrt{Y})^2$. Here $A_n^2 \leq k - c'_\eta$ for some $c'_\eta > 0$, since A^2 is continuous on the closed simplex and equals k only at the center. There are at most N^s bins, so they contribute $O(z^{2k}e^{(k-1-c'_\eta)z}) = o(e^{(k-1)z})$. This proves (2).

Step 5. Part (3). Equate the two expressions in (2) and let $N \rightarrow \infty$. This gives $k = D_k(2\pi)^{s/2}(\det \Sigma_k)^{1/2}$, that is, $D_k = k \det(2\pi \Sigma_k)^{-1/2}$. As a check, $\det((k^2/2)(I + J)) = (k^2/2)^s k$, which gives $D_k = k^{k+1/2}(4\pi)^{-s/2}$ again. $\square$

Proof of Theorem 3.8. At $p = 1/d$ the chain is a sequence of independent uniform states, and by the multinomial formula the move multiplies the probability by $n_b/(n_a + 1) > 1$. For $1/d < p < 1$ we have $\sigma > 0$ and $v = r/\sigma > 0$.

Step 1. An integral form. Writing $M! = \int_0^\infty e^{-t} t^M dt$ in the uniform case of Proposition 2.2(a) gives

$$P_N(n) = \frac{\sigma^{N-1}}{dv} \int_0^\infty e^{-t} \prod_i B_{n_i}(vt) dt, \quad B_n(y) = \sum_{m=1}^n \binom{n-1}{m-1} \frac{y^m}{m!}, \quad (17)$$

where the product runs over the positive coordinates.

Step 2. Log-concavity. We show that for each $y > 0$ the sequence $B_n(y)$, $n \geq 1$, is strictly log-concave. Write

$$\frac{B_n(y)}{y} = b_{n-1}, \quad b_j = \sum_{i=0}^j \binom{j}{i} a_i, \quad a_i = \frac{y^i}{(i+1)!}, \quad c_j = \sum_{i=0}^j \binom{j}{i} a_{i+1}.$$

Pascal's identity gives $b_{j+1} = b_j + c_j$. The ratio c_j/b_j is the mean of the strictly decreasing sequence $f_i = a_{i+1}/a_i = y/(i+2)$ under the weights $\lambda_i = \binom{j}{i} a_i$, $0 \leq i \leq j$, normalized. Let $\lambda'_i = \binom{j+1}{i} a_i$, $0 \leq i \leq j+1$, be the weights for $j+1$, and put $\lambda_{j+1} = 0$. On $0 \leq i \leq j$ the ratio $\lambda'_i/\lambda_i = (j+1)/(j+1-i)$ increases in i . So for $i < l$ we have $\lambda'_i \lambda_l - \lambda'_l \lambda_i \leq 0$, with strict inequality when $l = j+1$. Hence

$$\sum_i \lambda'_i f_i \sum_l \lambda_l - \sum_i \lambda'_i \sum_l \lambda_l f_l = \sum_{i < l} (\lambda'_i \lambda_l - \lambda'_l \lambda_i) (f_i - f_l) < 0,$$

since $f_i > f_l$ for $i < l$. So the mean decreases strictly, $c_{j+1}/b_{j+1} < c_j/b_j$, and $b_{j+1}/b_j = 1 + c_j/b_j$ decreases strictly with j . This is the strict log-concavity. Since $B_n(y) = (y/n)L_{n-1}^{(1)}(-y)$, it is a Turán-type inequality for the Laguerre polynomials $L_{n-1}^{(1)}(-y)$.

Step 3. The transfer. Let $b \geq a + 2$. It follows that $B_{a+1}(y)/B_a(y) > B_b(y)/B_{b-1}(y)$, that is, $B_{a+1}(y)B_{b-1}(y) > B_a(y)B_b(y)$. Apply this for $y = vt > 0$ in (17) and integrate. So the move strictly increases $P_N(n)$. Repeated moves reach a balanced bin from any bin with the same support, and all balanced bins with this support have the same probability by symmetry. This proves the theorem. $\square$

Proof of Corollary 3.9. For large N , $T \leq CL$ forces $r \rightarrow 0$ and so $p > 1/d$. By Theorem 3.8 every global mode is a vertex or a balanced bin, and all balanced bins with the same number of positive coordinates have the same probability. So we compare the ratios $C_{N,k}/V_N = S_n(u)$, $2 \leq k \leq d$, with 1 and with each other. Fix $0 < c < 1$.

Case $T \leq cL$. The ratio $S_n(u)$ increases with u , and u increases with T . So it is at most its value at $T = cL$, which is $\exp(-(k-1)(1-c)L + O(\log L)) \rightarrow 0$ by Theorem 3.1(i). So the vertices are the only modes, and $p > p_{N,d}$.

Case $cL \leq T \leq CL$. Theorem 3.1(i) gives $\log(C_{N,k}/V_N) = (k-1)(\Psi_N(T) - a_k) + o(1)$, uniformly. Fix $\delta > 0$, with $\delta < (a_{d-1} - a_d)/2$ when $d \geq 3$. This is possible since $a_2 > \dots > a_d$. If

$\Psi_N(T) \leq a_d - \delta$, every face ratio is at most $e^{-(k-1)\delta+o(1)} < 1$, so only the vertices are modes. If $\Psi_N(T) \geq a_d + \delta$, the full center beats the vertices, and it beats a balanced bin with $2 \leq k < d$ positive coordinates by

$$(d-k)(\Psi_N(T) - a_d) + (k-1)(a_k - a_d) + o(1) > 0.$$

If $|\Psi_N(T) - a_d| < \delta$, a proper face with $k \geq 2$ states has $(k-1)(\Psi_N(T) - a_k) + o(1) \leq -(k-1)\delta + o(1) < 0$, since $a_k - a_d > 2\delta$. So it loses to the vertices, and the comparison of the vertices with the full centers is decided by the unique crossing $p_{N,d}$ of Theorem 3.2. In each subcase the conclusion agrees with (i) to (iii).

The last claim follows from (5) and $p_{N,k} = 1 - (d-1)T_{N,k}/N$, since $a_2 > \dots > a_d$. $\square$

B Proofs for Sections 4 and 5

B.1 The start weights

We prove Lemma 4.1 and Proposition 4.3. Throughout $1/d < p < 1$, so $\sigma = p - r > 0$ and $v = r/\sigma = u/(1-u)$. The refresh form in Proposition 2.2(a) is linear in μ , and with $\mu = \delta_a$ it gives $p^{N-1}S_n^{(a)}(u)$. So, with sums over $1 \leq m_b \leq n_b$ for $b \in F$ and $M = \sum_b m_b$,

$$\omega_a(n) = \mathbb{E}_w \left[\frac{m_a}{M} \right], \quad w(m) \propto \frac{M!}{\prod_b m_b!} v^M \prod_{b \in F} \binom{n_b - 1}{m_b - 1}. \quad (18)$$

So $\omega_a(n)$ is the mean fraction of the blocks of the refresh form that carry state a .

Proof of Lemma 4.1. (1) Exchanging the letters a and b is a bijection from the words with counts n that start with a to those that start with b . It keeps the counts, since $n_a = n_b$, and the number of changes, so $S_n^{(a)} = S_n^{(b)}$. If k divides N , a balanced n has equal coordinates, and the k weights are equal and sum to 1.

(2) We may take $N \geq N_0$ so large that $p > 1/d$ and $\sigma \geq \frac{1}{2}$. Let n be balanced and $\nu = \lfloor N/k \rfloor$, so each n_b is ν or $\nu + 1$, and $\nu \geq N/(2k)$.

Step 1. Reduction to two values. By (1) the weights take a value ω_+ where $n_b = \nu + 1$ and a value ω_- where $n_b = \nu$. We show that $0 \leq \omega_+ - \omega_- \leq \delta$ with $\delta = \frac{2}{\nu}(\mathbb{E}_w M + \mathbb{E}_w M^2)$. Since the weights average to $1/k$, each is then within δ of $1/k$, and $|\sum_a \mu_a \omega_a(n) - \mu(F)/k| \leq \mu(F)\delta$.

Step 2. Pairing. Let $n_a = \nu + 1$ and $n_b = \nu$, and let m' be m with m_a and m_b exchanged, with $w(m') = 0$ if m' is not admissible. All factors of $w(m)$ other than $\binom{\nu}{m_a-1} \binom{\nu-1}{m_b-1}$ are symmetric in m_a and m_b . From $\binom{\nu}{j} / \binom{\nu-1}{j} = \nu/(\nu-j)$ we get $w(m') = w(m)(\nu - m_a + 1)/(\nu - m_b + 1)$ when $m_a \leq \nu$, and $w(m') = 0$ when $m_a = \nu + 1$. In both cases $w(m) - w(m') = w(m)(m_a - m_b)/(\nu - m_b + 1)$ when $m_a > m_b$. The map $m \mapsto m'$ matches the admissible m with $m_a < m_b$ to the admissible m with $m_b < m_a \leq \nu$. Pairing m with m' in (18) gives, with $Z = \sum_m w(m)$,

$$\omega_a - \omega_b = \frac{1}{Z} \sum_{m_a > m_b} (w(m) - w(m')) \frac{m_a - m_b}{M} = \frac{1}{Z} \sum_{m_a > m_b} w(m) \frac{(m_a - m_b)^2}{M(\nu - m_b + 1)} \geq 0.$$

Here $(m_a - m_b)^2/M \leq M$. If $m_b \leq \nu/2$ the last factor is at most $2/\nu$. If $m_b > \nu/2$, then $M > \nu/2$ and the term is at most $w(m)M < w(m)2M^2/\nu$. This proves the bound δ .

Step 3. The moments. It remains to bound $\mathbb{E}_w M$ and $\mathbb{E}_w M^2$. Since $(m-1)!(\frac{n_b-1}{m-1})/n_b^{m-1} = \prod_{j < m} (1 - j/n_b)_+$, we have $w(m) \propto c(m)A(m)$ for all m with every $m_b \geq 1$, where

$$c(m) = M! \prod_b \frac{(n_b v)^{m_b}}{m_b! (m_b - 1)!}, \quad A(m) = \prod_b \prod_{j < m_b} \left(1 - \frac{j}{n_b}\right)_+ \in [0, 1],$$

and Lemma A.2 gives $1 - A(m) \leq \sum_b m_b^2 / (2n_b) \leq M^2 / \nu$. Write $M! = \int_0^\infty e^{-t} t^M dt$. Since $\sum_{m \geq 1} y^m / (m!(m-1)!) = y\Phi(y)$, this gives $\mathbb{E}_c[(M+1)(M+2)] = \langle t^2 \rangle$, where $\langle \cdot \rangle$ is the mean under the density proportional to $e^{-t} t^k \prod_b \Phi(n_b v t)$, as in the proof of Theorem 6.2. The integration by parts there, with t^{k+2} in place of t^{k+1} , gives $\langle t^2 \rangle = (k+2)\langle t \rangle + \sum_b \langle t \psi(n_b v t) \rangle$. Put $c_1 = \sqrt{v} \sum_b \sqrt{n_b}$. That proof gives $\langle t \rangle \leq 2(k+1) + c_1^2$, and $\psi(x) \leq \sqrt{x}$ and $\langle t^{3/2} \rangle \leq \langle t^2 \rangle^{3/4}$ give $X = \langle t^2 \rangle \leq (k+2)(2k+2+c_1^2) + c_1 X^{3/4}$. If $c_1 X^{3/4} \geq X/2$, then $X \leq 16c_1^4$, and otherwise $X \leq 2(k+2)(2k+2+c_1^2)$. So $X \leq C_2(1 + c_1^2)^2$, where C_2 depends only on k . Here $c_1^2 \leq kNv = kT/\sigma \leq 2kT \leq 2kCL$, so $\mathbb{E}_c M^2 \leq \mathbb{E}_c[(M+1)(M+2)] \leq \nu/2$ for $N \geq N_0$. Since $0 \leq A \leq 1$ and $\mathbb{E}_c[A] \geq 1 - \mathbb{E}_c M^2 / \nu \geq \frac{1}{2}$, and $M \leq M^2$,

$$\mathbb{E}_w M \leq \mathbb{E}_w M^2 = \frac{\mathbb{E}_c[M^2 A]}{\mathbb{E}_c[A]} \leq \frac{\mathbb{E}_c M^2}{1 - \mathbb{E}_c M^2 / \nu} \leq 2C_2(1 + 2kT)^2.$$

With $\nu \geq N/(2k)$ this gives $\delta \leq C_1(1 + T)^2/N$. $\square$

Proof of Proposition 4.3. Take N large, so that $0 < u < 1$ and $z = Nu \in [cL, 2CL]$. Let $k = |F|$, put $s = k - 1$, $\Sigma = \sum_{b \in F} \sqrt{n_b} = A\sqrt{N}$ and $\lambda = u(A/\sqrt{\alpha_a} - 1)$. Note that $A/\sqrt{\alpha_a} - 1 = \sum_{b \neq a} \sqrt{\alpha_b/\alpha_a}$ lies in $[\sqrt{\varepsilon}, s/\sqrt{\varepsilon}]$. Let $j_1 = \lfloor N/\sqrt{L} \rfloor$, so that $j_1 \leq \varepsilon N/2$ for large N . We split the sum of Lemma 4.2 at j_1 . The constants below depend only on d, k, ε, c and C .

Step 1. One term. Let $1 \leq j \leq j_1$, and put $n' = n - je_a$, $N' = N - j$ and $\alpha' = n'/N'$. Then every $\alpha'_b \geq \varepsilon/2$, $|\alpha' - \alpha| \leq 2\sqrt{k}j/N$ and $N'u \in [z/2, z]$. Since $N' \geq N/2$, for large N both $N'u$ and z lie between $\frac{\varepsilon}{2} \log N'$ and $4C \log N'$. So Theorem 3.3, with $\varepsilon/2$ in place of ε and with the constants $\frac{\varepsilon}{2}$ and $4C$, applies at the totals N' and N with errors $O(1/L)$. The difference of its exponents is $(A(\alpha')^2 - 1)N'u - (A^2 - 1)z = u(\Sigma'^2 - \Sigma^2) + ju$, where $\Sigma' = \Sigma - \delta$ and $\delta = \sqrt{n_a} - \sqrt{n_a - j} = j/(\sqrt{n_a} + \sqrt{n_a - j})$. So $\delta \leq j/\sqrt{n_a}$, and

$$0 \leq \delta - \frac{j}{2\sqrt{n_a}} = \frac{j\delta}{2\sqrt{n_a}(\sqrt{n_a} + \sqrt{n_a - j})} \leq \frac{j^2}{2n_a^{3/2}}.$$

Since $\Sigma \leq \sqrt{kN}$ and $n_a \geq \varepsilon N$, this gives $u(\Sigma'^2 - \Sigma^2) = -2u\Sigma\delta + u\delta^2 = -juA/\sqrt{\alpha_a} + O(zj^2/N^2)$. Also $D(\alpha')/D(\alpha) = 1 + O(j/N)$, since D is smooth and positive on $\{\alpha_b \geq \varepsilon/2\}$, and the powers of $N'u$ and N' change by the factor $(1 - j/N)^{-s/2} = 1 + O(j/N)$. Hence there is C_2 with

$$S_{n-je_a}(u) = S_n(u) e^{-j\lambda + E_j}, \quad |E_j| \leq C_2 \left(\frac{1}{L} + \frac{zj^2}{N^2} + \frac{j}{N} \right). \quad (19)$$

Step 2. The main sum. For $j \leq j_1$ we have $zj^2/N^2 \leq ju/\sqrt{L}$ and $j/N \leq ju/(cL)$. So for large N we have $|E_j| \leq 1 + j\lambda/2$, and then $|e^{E_j} - 1| \leq e|E_j|e^{j\lambda/2}$. Put $\theta = \lambda/2$, so that $u\sqrt{\varepsilon}/2 \leq \theta \leq 1$. Since $1 - e^{-\theta} \geq \theta/2$, we have $\sum_{j \geq 1} j^m e^{-j\theta} \leq 16\theta^{-m-1}$ for $m = 0, 1, 2$. Hence

$$u \sum_{j \leq j_1} (1 - u)^{j-1} e^{-j\lambda} |e^{E_j} - 1| \leq eC_2 u \sum_{j \geq 1} e^{-j\theta} \left(\frac{1}{L} + \frac{zj^2}{N^2} + \frac{j}{N} \right) = O\left(\frac{u}{L\theta} + \frac{uz}{N^2\theta^3} + \frac{u}{N\theta^2} \right),$$

which is $O(1/L)$. The geometric sum is $u \sum_{j \geq 1} (1-u)^{j-1} e^{-j\lambda} = u/(e^\lambda - 1 + u)$. Since $\lambda = O(u)$ and $\lambda + u = uA/\sqrt{\alpha_a}$, it equals $(\sqrt{\alpha_a}/A)(1 + O(u))$. Its terms with $j > j_1$ add up to at most $u \sum_{j > j_1} e^{-j\lambda} \leq (2u/\lambda)e^{-j_1\lambda}$, which is $O(e^{-c\sqrt{\varepsilon}L/2})$. With (19) we get

$$u \sum_{j \leq j_1} (1-u)^{j-1} S_{n-je_a}(u) = S_n(u) \left(\frac{\sqrt{\alpha_a}}{A} + O\left(\frac{1}{L}\right) \right).$$

Step 3. The other terms. The map in Step 2 of the proof of the exact assertions in Theorem 3.2, which adds a copy of a just before its first occurrence, shows that $S_m(u) \leq S_{m+e_a}(u)$ for $u > 0$ whenever $m_a \geq 1$. So $S_{n-je_a}(u) \leq S_{n-j_1e_a}(u)$ for $j_1 < j < n_a$, and by (19) at $j = j_1$ these terms add up to at most $S_n(u)e^{-j_1\lambda+E_{j_1}}$, since $u \sum_j (1-u)^{j-1} \leq 1$. Here $zj_1^2/N^2 \leq z/L \leq 2C$ and $j_1/N \leq 1$, so $|E_{j_1}| \leq C_2(1/L + 2C + 1) = O(1)$. Also $j_1\lambda \geq (N/\sqrt{L} - 1)u\sqrt{\varepsilon} \geq c\sqrt{\varepsilon}L/2$ for large N . So these terms are $O(S_n(u)e^{-c\sqrt{\varepsilon}L/2}) = O(S_n(u)/L)$. The term $j = n_a$ is at most $u(1-u)^{n_a-1}S_{n''}(u) \leq 2ue^{-\alpha_a z}S_{n''}(u)$, where n'' has the $k-1$ coordinates n_b with $b \neq a$, and total $N'' = N(1-\alpha_a)$. Theorem 3.3 gives $S_{n''}(u) \geq c_3 z^{s/2} N^{-s} e^{(A^2-1)z}$ with $c_3 > 0$. If $k = 2$, then $S_{n''} = 1$ and $A^2 - 1 = 2\sqrt{\alpha_1\alpha_2} \geq 2\varepsilon$, so the term is $O(S_n(u)z^{1/2}e^{-2\varepsilon z})$. If $k \geq 3$, the fractions n_b/N'' are at least ε and $N''u \in [\varepsilon z, z]$. Since $(\sum_{b \neq a} \sqrt{n_b})^2 = N(A - \sqrt{\alpha_a})^2$, Theorem 3.3 on $k-1$ states, with the constants $\frac{\varepsilon c}{2}$ and $4C$, gives $S_{n''}(u) \leq C_3 z^{(s-1)/2} N''^{1-s} e^{z[(A-\sqrt{\alpha_a})^2-1+\alpha_a]}$. So the term is $O(S_n(u)z^{1/2}e^{-z\sqrt{\alpha_a}(2A-\sqrt{\alpha_a})}) = O(S_n(u)z^{1/2}e^{-\varepsilon z})$. In both cases it is $O(S_n(u)/L)$.

Adding the three steps and dividing by $S_n(u)$ gives $\omega_a(n) = S_n^{(a)}(u)/S_n(u) = \sqrt{\alpha_a}/A + O(1/L)$. $\square$

B.2 Gaussian confinement

The next lemma gives the Gaussian profile of the ratio to a vertex near the center of a face. We use it for Theorems 4.6 and 5.2.

Lemma B.1 (Gaussian confinement). *Fix a support S with $|S| = k \geq 2$ and a compact set $\mathcal{J}$ of values of t , put $T = L - \frac{1}{2} \log L + t$, and let $u = r/p$ be the odds at this T . For a positive count vector n on S put $y = \sqrt{L}(n/N - \frac{1}{k}\mathbf{1})$, a vector in the hyperplane $\{y \in \mathbb{R}^S : \sum_i y_i = 0\}$. The terms $o(1)$ below tend to 0 as $N \rightarrow \infty$.*

(a) *Uniformly for $t \in \mathcal{J}$ and y in a compact subset of this hyperplane,*

$$\log S_n(u) = (k-1)(t - a_k) - \frac{k^2}{4}|y|^2 + o(1).$$

(b) *For every $R > 0$, uniformly for $t \in \mathcal{J}$ and all n with $|y| > R$,*

$$\log S_n(u) \leq (k-1)(t - a_k) - \frac{k^2}{4}R^2 + o(1).$$

Proof. (a) Put $\alpha = n/N = \frac{1}{k}\mathbf{1} + y/\sqrt{L}$. For $|y|$ bounded, $\alpha_i \geq 1/(2k)$ for large N , and Taylor expansion of $\sqrt{1/k+x} = k^{-1/2} + \frac{\sqrt{k}}{2}x - \frac{k^{3/2}}{8}x^2 + O(x^3)$, with $\sum_i y_i = 0$, gives

$$\left(\sum_i \sqrt{\alpha_i} \right)^2 = k - \frac{k^2}{4L}|y|^2 + O(L^{-3/2}), \quad D(\alpha) = D_k(1 + o(1)).$$

Insert these into Theorem 3.3, with $\varepsilon = 1/(2k)$, $c = \frac{1}{2}$, $C = 2$ and $z = Nu = T/p$, where $z - T = O(L^2/N)$. Then $(A^2 - 1)z = (k - 1)z - \frac{k^2}{4}|y|^2 + o(1)$ and $z + \frac{1}{2}\log z - L = t + \frac{1}{2}\log(z/L) + o(1) = t + o(1)$. Since $\log D_k = -(k - 1)a_k$, this gives (a).

(b) For large N we have $p > 1/d$, since $T = O(L)$. Starting from n , move one unit from a coordinate b to a coordinate a with $n_b \geq n_a + 2$, and repeat. The moves keep the support and end at a balanced vector, and by Theorem 3.8 each move strictly increases $S_n(u)$. A move changes $|y|^2$ by $\frac{2L}{N^2}(n_a - n_b + 1) < 0$ and changes y by a vector of length $\sqrt{2L}/N$. The balanced end has $|y| = O(\sqrt{L}/N)$. So if $|y| > R$ at the start, the first vector on the path with $|y| \leq R$ has $|y| \geq R - \sqrt{2L}/N = R - o(1)$, and by (a), applied on the compact set $\{|y| \leq R\}$, its value of $\log S_n$ is at most $(k - 1)(t - a_k) - k^2 R^2/4 + o(1)$, uniformly. The starting vector has a smaller value. $\square$

B.3 Face maxima with a start

Let F be a face with $k \geq 2$ states and put $s = k - 1$. For $1 \leq j \leq d$ and $M \geq j$ let $\bar{S}_{M,j}(u)$ be the value of $S_m(u)$ at a balanced vector m with j positive coordinates and total M . So $\bar{S}_{N,k} = C_{N,k}/V_N$ and $\bar{S}_{M,1} = 1$. We use two facts. First, if $0 < u \leq 1$, then $S_m(u) \leq \bar{S}_{M,j}(u)$ for every positive vector m with j coordinates and total M . This is Theorem 3.8, since $S_m(u) = P_M(m)/V_M$ and $u \leq 1$ means $p \geq 1/d$. Second, if $u > 0$, then $\bar{S}_{M,j}(u) \leq \bar{S}_{M+1,j}(u)$, by Step 2 of the proof of the exact assertions in Theorem 3.2.

Lemma B.2 (the start weights from above). *Fix $0 < c < C$. There is C_4 such that for large N , uniformly for $cL \leq T \leq CL$,*

$$u \sum_{1 \leq j \leq N/2} (1 - u)^{j-1} \bar{S}_{N-j,k}(u) \leq \frac{1}{k} \left(1 + \frac{C_4}{L}\right) \bar{S}_{N,k}(u), \quad (20)$$

and $kS_n^{(a)}(u) \leq (1 + C_4/L) \bar{S}_{N,k}(u)$ for every positive count vector n on F and every $a \in F$.

Proof. Take N so large that $p > 1/d$. Then $0 < u < 1$, and $z = Nu = T/p$ lies in $[cL, 2CL]$.

Step 1. The sum. Let $1 \leq j \leq N/2$ and $M = N - j$. Then $M \geq N/2$ and $Mu = z(1 - j/N) \in [z/2, z]$, so for large N both Mu and z lie between $\frac{c}{2} \log M$ and $4C \log M$. So (6) applies at the totals M and N , with the constants $\frac{c}{2}$ and $4C$, and its errors are $O(1/L)$. Since $Mu - z = -ju$, it gives

$$\frac{\bar{S}_{M,k}(u)}{\bar{S}_{N,k}(u)} = \left(1 - \frac{j}{N}\right)^{-s/2} e^{-sj u} \left(1 + O\left(\frac{1}{L}\right)\right).$$

For $0 \leq x \leq \frac{1}{2}$ we have $\log(1 - x) \geq -2x$, since the left side is concave and the inequality holds at both ends. So $(1 - j/N)^{-s/2} \leq e^{sj/N}$. Put $q = ku - s/N = ku(1 - s/(kz))$, which is positive. With $(1 - u)^{j-1} \leq e^{-(j-1)u}$ and $\sum_{j \geq 1} e^{-jq} = 1/(e^q - 1) \leq 1/q$, the left side of (20) is at most

$$\left(1 + O\left(\frac{1}{L}\right)\right) \bar{S}_{N,k}(u) u e^u \sum_{j \geq 1} e^{-jq} \leq \left(1 + O\left(\frac{1}{L}\right)\right) \bar{S}_{N,k}(u) \frac{u e^u}{q}.$$

Here $u e^u/q = e^u/(k(1 - s/(kz))) = (1 + O(1/L))/k$, since $u = O(L/N)$ and $z \geq cL$. This proves (20).

Step 2. The bound. Apply the first fact to the vectors $n - j e_a$ in Lemma 4.2. They have k positive coordinates for $j < n_a$ and $k - 1$ for $j = n_a$. With $(1 - u)^{n_a-1} \leq 1$ and the second fact, this gives

$$S_n^{(a)}(u) \leq u \sum_{1 \leq j \leq N/2} (1 - u)^{j-1} \bar{S}_{N-j,k}(u) + \bar{S}_{N,k}(u) \sum_{j > N/2} u(1 - u)^{j-1} + u \bar{S}_{N,k-1}(u).$$

The first term is bounded by (20). In the second, $\sum_{j>N/2} u(1-u)^{j-1} \leq (1-u)^{(N-1)/2} \leq 2e^{-z/2} \leq 2N^{-c/2}$. For the third, Theorem 3.1(i), for k and for $k-1$ when $k \geq 3$, gives $\bar{S}_{N,k-1}(u)/\bar{S}_{N,k}(u) = O(e^{-\Psi_N(T)})$. This also holds for $k=2$, since $\bar{S}_{N,1} = 1$. Here $ue^{-\Psi_N(T)} = zT^{-1/2}e^{-T} = O(L^{1/2}N^{-c})$. So the second and third terms are $O(\bar{S}_{N,k}(u)/L)$, which proves the bound. $\square$

Proof of Theorem 4.6. (1) Let n have support F . By Proposition 2.2(b) and Lemma B.2,

$$\frac{P_N^\mu(n)}{p^{N-1}} = \sum_{a \in F} \mu_a S_n^{(a)}(u) \leq \mu(F) \max_{a \in F} S_n^{(a)}(u) \leq \frac{\mu(F)}{k} \left(1 + \frac{C_4}{L}\right) \bar{S}_{N,k}(u).$$

By Theorem 4.4(2), a balanced bin of F has $P_N^\mu(n)/p^{N-1} = \frac{\mu(F)}{k} \bar{S}_{N,k}(u)(1 + O(L^2/N))$. Since $p^{N-1} \bar{S}_{N,k}(u) = dC_{N,k}$, this proves the first display. The second follows from Theorem 3.1(iii), which gives $\log \bar{S}_{N,k}(u) = (k-1)(t - a_k) + o(1)$.

(2) Fix $\mathcal{J}$, $R > 0$ and $K \geq 1$. Let n have support F and $|y| > R$, let $a \in F$, and put $j_K = \lfloor KN/L \rfloor$. In the window $z/L \rightarrow 1$, so for large N we have $j_K \leq N/2$ and $uj_K \geq K/2$. As in Step 2 of the proof of Lemma B.2, but with the sum split at j_K ,

$$S_n^{(a)}(u) \leq u \sum_{j \leq j_K, j < n_a} (1-u)^{j-1} S_{n-je_a}(u) + (e^{-K/2} + O(L^{-1})) \bar{S}_{N,k}(u).$$

Now let $j \leq j_K$ and $j < n_a$. Put $n' = n - je_a$, $N' = N - j$, $L' = \log N'$ and $y' = \sqrt{L'}(n'/N' - \frac{1}{k}\mathbf{1})$. The chain of length N' with the same p has $T' = N'r$, the same u , and the window coordinate $t' = T' - L' + \frac{1}{2} \log L'$. Since $T - T' = Tj/N \leq 2K$ and $0 \leq L - L' \leq 2j/N \leq 2K/L$, the value t' lies in a compact set $\mathcal{J}'$ that depends only on $\mathcal{J}$ and K . Each coordinate of $n'/N' - n/N$ has absolute value at most $j/N' \leq 2j/N$, so $|n'/N' - n/N| \leq 2\sqrt{k}K/L$. With $|y| \leq 2\sqrt{L}$ and $\sqrt{L'/L} \geq 1 - 2K/L^2$ this gives $|y'| \geq |y| - O(L^{-1/2}) > R/2$ for large N . So Lemma B.1(b) at length N' , with $\mathcal{J}'$ and $R/2$, and Theorem 3.1(iii) at length N' give $S_{n'}(u) \leq e^{-k^2 R^2/16 + o(1)} \bar{S}_{N',k}(u)$, uniformly. With (20) we get

$$kS_n^{(a)}(u) \leq (e^{-k^2 R^2/16} + ke^{-K/2} + o(1)) \bar{S}_{N,k}(u).$$

Choose K with $ke^{-K/2} < \frac{1}{2}(1 - e^{-k^2 R^2/16})$. Then, as in (1), there is $\delta > 0$ such that for large N every bin with support F and $|y| > R$ has $P_N^\mu(n) \leq (1 - \delta) \frac{\mu(F)}{k} p^{N-1} \bar{S}_{N,k}(u)$, while a balanced bin has $1 - o(1)$ times the same value. So no maximizer has $|y| > R$. $\square$

Proof of Corollary 4.7. By Proposition 2.2(b), a vertex i has $P_N^\mu(Ne_i) = \mu_i p^{N-1}$, and every bin on a face S with $\mu(S) = 0$ has probability 0. For a face S with $|S| = k \geq 2$ and $\mu(S) > 0$, Theorem 4.6(1) gives the best bin of S the value $\log \frac{\mu(S)}{k} + (k-1)(t - a_k) + o(1)$ of $\log(P_N^\mu/p^{N-1})$, uniformly for $t \in \mathcal{J}$. Among the faces of size k the largest $\mu(S)$ is M_k . So this value is $\ell_k^\mu(t) + o(1)$ when $\mu(S) = M_k$, and it is at most $\ell_k^\mu(t) - \delta_1 + o(1)$ when $0 < \mu(S) < M_k$, where $\delta_1 > 0$ depends only on μ . The same holds for the vertices, with ℓ_1^μ and without the term $o(1)$. Since $\mathcal{J}$ is compact and the lines are continuous, there is $\delta_2 > 0$ with $\ell_{k^*}^\mu(t) \geq \ell_k^\mu(t) + \delta_2$ for all $t \in \mathcal{J}$ and $k \neq k^*$. There are finitely many faces. So for large N and all $t \in \mathcal{J}$, the best bin of a face of size k^* with $\mu(S) = M_{k^*}$ is more likely than every bin on every face that does not have size k^* and mass $\mu(S) = M_{k^*}$. This proves the first claim and the claim for $k^* = 1$. If μ is constant on S , then $P_N^\mu(n) = \mu_a p^{N-1} S_n(u)$ for $a \in S$ and every n with support S , and $S_n(u)$ is symmetric in the coordinates. Since $p > 1/d$ for large N , Theorem 3.8 gives the claim on balanced bins. The last claim is Theorem 4.6(2), since a global mode maximizes P_N^μ among the bins with its support, and each coordinate of y is at most $|y|$ in absolute value. $\square$

B.4 The field

Proof of Theorem 5.2. Fix compact sets of t and h , and a support S with $|S| = k \geq 2$.

Step 1. The best bin on S . Relative to V_N , a bin with support S has weight $e^{h \cdot \alpha} S_n(u)$ with $\alpha = n/N$. Choose R with $k^2 R^2/4 > \max_i h_i - \bar{h}_S + 1$ on the compact set. Since $h \cdot \alpha \leq \max_{i \in S} h_i$, Lemma B.1(b) shows that bins with $|y| > R$ have log weight at most $\ell_S(t, h) - 1 + o(1)$. On $|y| \leq R$ we have $h \cdot \alpha = \bar{h}_S + O(R/\sqrt{L})$, and Lemma B.1(a) gives the log weight $\ell_S(t, h) - \frac{k^2}{4}|y|^2 + o(1)$. A balanced bin has $|y| = O(\sqrt{L}/N)$. So the largest weight on S is $e^{\ell_S(t, h) + o(1)}$, while a vertex i has weight exactly e^{h_i} .

Step 2. The modes. There are finitely many supports. This proves (10), and a positive gap to a lower score excludes that support for large N . If the maximizers on a selected support had $|y| \geq \eta > 0$ along a subsequence, their log weight would be at most $\ell_S - k^2 \eta^2/4 + o(1)$, below that of a balanced bin. This proves the location statement.

Step 3. The lines. Among supports of size k the largest $\bar{h}_S$ is the average of the k largest fields, which gives (11). If k_1 wins at t_1 and k_2 at $t_2 > t_1$, adding the two comparisons gives $(k_1 - k_2)(t_2 - t_1) \leq 0$. So the winning size is nondecreasing away from ties.

Step 4. Every intermediate size. To realize a given $2 \leq k < d$, set $h_1 = \dots = h_k = 0$ and $h_{k+1} = \dots = h_d = -H$, and take $t > a_k$ so close to a_k that $t < a_{k-1}$ if $k > 2$. Then the k -face score is positive, the largest vertex score is 0, and every face with $2 \leq j < k$ states scores at most $(j-1)(t - a_j) < 0$. A face with $j > k$ states scores at most $(j-1)(t - a_j) - H(j-k)/j$, and another face with k states at most $(k-1)(t - a_k) - H/k$, so both lose once H is large. The comparisons are strict, so they hold on an open interval of t , and small changes make the fields strictly ordered without removing the phase. $\square$

Proof of Corollary 5.4. Since $\frac{1}{k} \sum_{i=1}^k (i-1)^2 = (k-1)(2k-1)/6$, the lines (11) are $\ell_k(t) = (k-1)t + c_k - H(k-1)(2k-1)/6$, also for $k=1$, and $\ell_{k+1}(t) - \ell_k(t) = t - t_k$. Extend c_k to the smooth function $c(x) = (x + \frac{1}{2}) \log x - \frac{x-1}{2} \log(4\pi)$ for $x \geq 1$. Then $0 < c''(x) = (x - \frac{1}{2})/x^2 \leq \frac{1}{2}$, and integrating twice over unit intervals gives $c_{k+2} - 2c_{k+1} + c_k \leq \frac{1}{2}$. Hence

$$t_{k+1} - t_k = \frac{2H}{3} - (c_{k+2} - 2c_{k+1} + c_k) > 0.$$

For $t_{k-1} < t < t_k$, summing the differences $\ell_{j+1}(t) - \ell_j(t) = t - t_j$ shows that line k strictly exceeds every other line. The fields are strictly decreasing, so the unique best support of size k is $\{1, \dots, k\}$, and Theorem 5.2 gives the conclusion for finite N . $\square$

C The majorant and the crossover at $N/(\log N)^2$

This appendix proves Lemma 6.1, Theorem 6.2 and the results behind Remark 6.3. The start is uniform.

C.1 The majorant

Proof of Lemma 6.1. By definition, $B_a(w)/w = \sum_{j=0}^{a-1} \binom{a-1}{j} w^j / (j+1)!$. So the coefficient of $(aw)^j / [j!(j+1)!]$ in $B_a(w)/w$ is $\binom{a-1}{j} j! / a^j = \prod_{h=1}^j (1 - h/a)_+$, also for $j \geq a$, where it is 0. By Lemma A.2 its deficit from 1 lies between 0 and $\sum_{h \leq j} h/a = j(j+1)/(2a)$. Since

$\sum_j j(j+1)x^j/[j!(j+1)!] = x\Phi(x)$ (Lemma A.1(i)), we get

$$0 \leq \Phi(aw) - \frac{B_a(w)}{w} \leq \frac{1}{2a} \sum_{j \geq 0} \frac{j(j+1)(aw)^j}{j!(j+1)!} = \frac{w}{2} \Phi(aw). \quad \square$$

Proof of Theorem 6.2. Step 1. An integral form. Let $0 < u < 1$ and $v = u/(1-u)$. Put $M! = \int_0^\infty e^{-t} t^M dt$ into (3) and sum over each m_i separately. All terms are nonnegative, so

$$S_n(u) = \frac{(1-u)^{N-1}}{v} \int_0^\infty e^{-t} \prod_{i=1}^k B_{n_i}(vt) dt, \quad (21)$$

with B_a as in Lemma 6.1.

Step 2. Comparison of the factors. For $t > 0$ put $f_i(t) = B_{n_i}(vt)/(vt\Phi(n_i vt)) \in [0, 1]$. By Lemma 6.1 we have $f_i(t) = (1 - \theta_i)_+$ with $0 \leq \theta_i \leq vt/2$, so Lemma A.2 gives $0 \leq 1 - \prod_i f_i(t) \leq kv t/2$. Write $\langle g \rangle$ for the mean of $g(t)$ under the probability density on $(0, \infty)$ proportional to $e^{-t} t^k \prod_i \Phi(n_i vt)$. Then (21) reads $S_n(u) = \mathcal{Q}_n(u) \langle \prod_i f_i \rangle$, and so

$$0 \leq 1 - \frac{S_n(u)}{\mathcal{Q}_n(u)} = \left\langle 1 - \prod_i f_i \right\rangle \leq \frac{kv}{2} \langle t \rangle.$$

The bound $\Phi(x) \leq e^{2\sqrt{x}}$ of Section 2 bounds the density by a polynomial times $\exp(-t + 2\sqrt{vt} \sum_i \sqrt{n_i})$, so all moments below exist.

Step 3. The mean of t . We use $\psi(x) = x\Phi'(x)/\Phi(x)$ from Lemma A.1. Since $\frac{d}{dt} \log \Phi(n_i vt) = \psi(n_i vt)/t$,

$$\frac{d}{dt} \left[e^{-t} t^{k+1} \prod_i \Phi(n_i vt) \right] = e^{-t} t^k \prod_i \Phi(n_i vt) \left(k + 1 - t + \sum_i \psi(n_i vt) \right).$$

The function in square brackets vanishes at $t = 0$ and tends to 0 as $t \rightarrow \infty$, so integration gives $\langle t \rangle = k + 1 + \sum_i \langle \psi(n_i vt) \rangle$. By Lemma A.1, $\psi(x) \leq \sqrt{x}$. With $\langle \sqrt{t} \rangle \leq \sqrt{\langle t \rangle}$ and $\Gamma = \sqrt{v} \sum_i \sqrt{n_i}$ this gives $\langle t \rangle \leq k + 1 + \Gamma \sqrt{\langle t \rangle}$. Since $\Gamma \sqrt{\langle t \rangle} \leq (\Gamma^2 + \langle t \rangle)/2$, we get $\langle t \rangle \leq 2(k + 1) + v(\sum_i \sqrt{n_i})^2$. This proves the first bound in (13), and Cauchy-Schwarz, $(\sum_i \sqrt{n_i})^2 \leq kN$, gives the second. $\square$

C.2 The crossover

Fix integers $k_1 \geq 2$ and $\ell \geq 0$, and put $k = k_1 + \ell$. We take k_1 large counts $n_1, \dots, n_{k_1}$ and ℓ rare counts $a_1, \dots, a_\ell$, all positive, and write $S_{(n,a)}$ for the ratio S of this count vector. For $\ell = 0$ there are no rare counts, and the sums and products over j below are empty. Put

$$N = \sum_i n_i + \sum_j a_j, \quad M = \sum_i n_i, \quad L = \log N, \quad \beta_i = \frac{n_i}{M}, \quad \rho_j = \frac{a_j L^2}{N}.$$

The β_i and ρ_j always refer to these actual counts. In this appendix

$$A = \sum_{i=1}^{k_1} \sqrt{\beta_i}, \quad g = A^2 - 1, \quad \eta = \frac{k_1 - 1}{2} + 2\ell, \quad c_* = \frac{k - 1}{g},$$

$$D_{k_1}(\beta) = \frac{A^{k_1/2+1} (\prod_i \beta_i)^{-3/4}}{(4\pi)^{(k_1-1)/2}}, \quad \gamma_0 = \frac{(2\eta + 1)(2\eta + 3)}{16A^2} - \frac{3}{16A} \sum_{i=1}^{k_1} \beta_i^{-1/2}.$$

The constant $D_{k_1}(\beta)$ is $D(\beta)$ of Theorem 3.3 on k_1 states, so it has the Kirchhoff form of Proposition 3.6, $D_{k_1}(\beta) = A^2(4\pi)^{-(k_1-1)/2} \sqrt{\text{tree}(\beta)} / \prod_i \beta_i$. We use $\psi(x) = x\Phi'(x)/\Phi(x)$ and its properties from Lemma A.1.

Theorem C.1 (a crossover between neighboring simplex strata). *Fix $\varepsilon > 0$ and $R < \infty$. Assume $\beta_i \geq \varepsilon$ and $0 < \rho_j \leq R$. For $z > 0$ put $x_j = A^2 a_j z^2 / N$ and*

$$\mathcal{E}(z) = -gz \sum_{j=1}^{\ell} \frac{a_j}{N} + \frac{1}{A^2 z} \left(\eta \sum_{j=1}^{\ell} \psi(x_j) + 2 \sum_{1 \leq i < j \leq \ell} \psi(x_i) \psi(x_j) \right) + \frac{\gamma_0}{z},$$

so that $\mathcal{E}(z) = \gamma_0/z$ when $\ell = 0$. Let $0 < c_0 < C_0 < \infty$ be fixed. Then, uniformly for such count vectors and $c_0 L \leq z \leq C_0 L$,

$$S_{(n,a)}(z/N) = D_{k_1}(\beta) A^{2\ell} \frac{z^\eta e^{gz}}{N^{k-1}} \prod_{j=1}^{\ell} \Phi(x_j) e^{\mathcal{E}(z)} \left(1 + O\left(L^{-2} + \frac{L^2}{N}\right) \right). \quad (22)$$

Also $|\mathcal{E}(z)| \leq C_1(1+R)/L$ there, where C_1 depends only on $\varepsilon, c_0, C_0, k_1$ and ℓ . So (22) also holds without the factor $e^{\mathcal{E}(z)}$ if the error is replaced by $O(L^{-1} + L^2/N)$.

For large N , the right side of (22) without its error factor equals 1 at exactly one ζ_N in $[(k-1)L/(2(k_1-1)), (k-1)L/\varepsilon]$. The unique positive solution $u_{n,a}$ of $S_{(n,a)}(u) = 1$ satisfies

$$Nu_{n,a} = \zeta_N + O\left(L^{-2} + \frac{L^2}{N}\right) \quad (23)$$

and

$$\begin{aligned} Nu_{n,a} = c_* L - \frac{\eta}{g} \log L - \frac{1}{g} \left\{ \log(D_{k_1}(\beta) A^{2\ell}) + \eta \log c_* + \sum_{j=1}^{\ell} \log \Phi(A^2 c_*^2 \rho_j) \right\} \\ + O\left(\frac{\log L}{L} + \frac{L^2}{N}\right). \end{aligned} \quad (24)$$

The constants in the errors of (22) depend only on $\varepsilon, R, k_1, \ell, c_0$ and C_0 . Those in (23) and (24), and the meaning of “large N ”, depend only on ε, R, k_1 and ℓ .

So each rare count (a state with few visits) multiplies the profile of the large counts by $A^2 z^2 / N$ and by a Bessel factor $\Phi(x_j)$. At $u = z/N$ we have $T = z + O(L^2/N)$, the logarithms of the right sides have bounded derivatives in z , and all errors contain L^2/N . So the theorem also holds with T in place of z and of $Nu_{n,a}$.

Proof. Let $u = z/N$ with $c_0 L \leq z \leq C_0 L$, and take N large. Put $v = u/(1-u)$, $Y = Mv$, $w_0 = A\sqrt{Y}$, $q_0 = \eta + 3/2 = 2\ell + 1 + k_1/2$, $s_a = \sum_j a_j/N$ and $\xi_j = a_j v w_0^2 = A^2 a_j M v^2$. Then $s_a \leq \ell R/L^2$ and $Y = z(1-s_a)/(1-z/N)$, so $Y \asymp L$. Also $\xi_j \leq X = 4k_1 C_0^2 R$. When $\ell = 0$ we have $M = N$ and $s_a = 0$, and the steps that involve the ξ_j and X are empty. Note that $A^2 = 1 + 2 \sum_{i < i'} \sqrt{\beta_i \beta_{i'}}$ lies in $[1 + 2\varepsilon, k_1]$.

Step 1. The Bessel integral. By Theorem 6.2, with relative error $O(v + Nv^2) = O(L^2/N)$, the ratio $S_{(n,a)}(u)$ equals $\mathcal{Q}_{(n,a)}(u)$. Writing $\Phi(\beta_i Y t) = I_1(2\sqrt{\beta_i Y t})/\sqrt{\beta_i Y t}$ for the large counts, this is

$$\frac{(1-u)^{N-1} v^{\ell-1}}{\prod_{i=1}^{k_1} n_i} \int_0^\infty e^{-t} t^\ell \prod_{i=1}^{k_1} \left[\sqrt{\beta_i Y t} I_1(2\sqrt{\beta_i Y t}) \right] \prod_{j=1}^{\ell} \Phi(a_j v t) dt. \quad (25)$$

Substitute $t = w^2$. The integrand becomes

$$2w^{2\ell+1}e^{-w^2} \prod_{i=1}^{k_1} \left[\sqrt{\beta_i Y} w I_1(2\sqrt{\beta_i Y} w) \right] \prod_{j=1}^{\ell} \Phi(a_j v w^2).$$

Put $F(w) = w^{q_0} \prod_j \Phi(a_j v w^2)$, so that $F(w_0) = A^{q_0} Y^{q_0/2} \prod_j \Phi(\xi_j)$.

Step 2. The main part. On $I = \{|w - w_0| \leq w_0/2\}$ the Bessel arguments $2\sqrt{\beta_i Y} w$ are at least $A\sqrt{\varepsilon} Y$. The expansion of I_1 in Section 2, and $-w^2 + 2A\sqrt{Y} w = A^2 Y - (w - w_0)^2$, then show that on I the integrand equals

$$\frac{2(\prod_i \beta_i)^{1/4} Y^{k_1/4}}{(4\pi)^{k_1/2}} e^{A^2 Y - (w - w_0)^2} F(w) \left(1 - \frac{b_Y}{w} + O(Y^{-2}) \right), \quad b_Y = \frac{3}{16\sqrt{Y}} \sum_i \beta_i^{-1/2}.$$

We apply Lemma A.3 with $c_j = a_j v$, once with $q = q_0$ to F and once with $q = q_0 - 1$ to $F(w)/w$ for the term b_Y/w . For the term $O(Y^{-2})$ we use its bound $F(w_0 + y) = O(F(w_0))$. Since $b_Y/w_0 = O(1/Y)$, the integral over I equals

$$\frac{2\sqrt{\pi}(\prod_i \beta_i)^{1/4} Y^{k_1/4}}{(4\pi)^{k_1/2}} e^{A^2 Y} F(w_0) \left(1 + \frac{\mu_{q_0}}{4A^2 Y} - \frac{3}{16AY} \sum_i \beta_i^{-1/2} + O(Y^{-2}) \right),$$

with μ_{q_0} as in the lemma.

Step 3. The tails. The part outside I is negligible. The bounds $I_1(x) \leq e^x$ and $\Phi(x) \leq e^{2\sqrt{x}}$ bound the integrand there by a polynomial in w and Y times $\exp(A^2 Y - (w - w_0)^2 + 2\delta w)$, where $\delta = \sum_j \sqrt{a_j v}$, and $\delta w_0 = \sum_j \sqrt{\xi_j}$ is bounded. Completing the square gives $-(w - w_0 - \delta)^2 + 2\delta w_0 + \delta^2$, and outside I we have $|w - w_0 - \delta| \geq 3w_0/8$ for large N . Hence this part is at most a power of Y times $e^{A^2 Y - 9A^2 Y/64}$, which is $O(L^{-2})$ relative to the main term.

Step 4. Collecting the constants. Since $v^{\ell-1}/\prod_i n_i = Y^{\ell-1} M^{-(k-1)}/\prod_i \beta_i$ and $\ell - 1 + k_1/4 + q_0/2 = \eta$, the constants collect to $D_{k_1}(\beta) A^{2\ell} Y^\eta M^{-(k-1)}$. Since $4q_0 - 6 = 4\eta$ and $q_0(q_0 - 1) = (2\eta + 1)(2\eta + 3)/4$, the formula for μ_q in Lemma A.3 gives

$$\frac{\mu_{q_0}}{4A^2} - \frac{3}{16A} \sum_i \beta_i^{-1/2} = \frac{1}{A^2} \sum_j \xi_j + \Theta(\xi), \quad \Theta(\xi) = \frac{1}{A^2} \left(\eta \sum_j \psi(\xi_j) + 2 \sum_{i < j} \psi(\xi_i) \psi(\xi_j) \right) + \gamma_0.$$

Note that $\sum_j \xi_j/(A^2 Y) = v \sum_j a_j$. Both $\sum_j \xi_j/A^2$ and $\Theta(\xi)$ are bounded, so the bracket in Step 2 equals $\exp(v \sum_j a_j + \Theta(\xi)/Y)(1 + O(Y^{-2}))$. We have proved

$$S_{(n,a)}(u) = D_{k_1}(\beta) A^{2\ell} \frac{Y^\eta}{M^{k-1}} (1-u)^{N-1} e^{v(A^2 M + \sum_j a_j)} \prod_j \Phi(\xi_j) e^{\Theta(\xi)/Y} \left(1 + O\left(L^{-2} + \frac{L^2}{N}\right) \right). \quad (26)$$

Step 5. From (M, Y) to (N, z) . Now use $u = z/N$. Since $A^2 M + \sum_j a_j = N(A^2 - g s_a)$ and $vN = z + O(L^2/N)$, the exponent is

$$v \left(A^2 M + \sum_j a_j \right) + (N-1) \log(1-u) = gz - gzs_a + O(L^2/N).$$

Also $Y^\eta M^{-(k-1)} = z^\eta N^{-(k-1)}(1 + O(L^{-2} + L/N))$ and $\xi_j = x_j(1 - s_a)(1 - z/N)^{-2} = x_j(1 + O(L^{-2} + L/N))$. Since $(\log \Phi)'(x) = \psi(x)/x \leq 1/2$, replacing ξ_j by x_j in Φ changes the product

by a relative $O(L^{-2} + L/N)$. Since $0 \leq \psi' \leq 1$, replacing $\Theta(\xi)/Y$ by $\Theta(x)/z$ changes it by $O(L^{-3} + 1/N)$. Finally $\mathcal{E}(z) = -gzs_a + \Theta(x)/z$, which proves (22).

Step 6. The bound on $\mathcal{E}$. Note that $\psi(x_j) \leq \sqrt{x_j}$, $2\psi(x_i)\psi(x_j) \leq \psi(x_i)^2 + \psi(x_j)^2 \leq x_i + x_j$, and $A \geq 1$. So $|\mathcal{E}(z)| \leq gzs_a + (\eta \sum_j \sqrt{x_j} + (\ell - 1) \sum_j x_j + |\gamma_0|)/z$. Here $gzs_a \leq (k_1 - 1)C_0\ell R/L$, $x_j = A^2\rho_j(z/L)^2 \leq k_1 C_0^2 R$, $z \geq c_0 L$ and $\sqrt{R} \leq 1 + R$, and $|\gamma_0|$ is bounded in terms of ε , k_1 and ℓ .

Step 7. The crossing. Apply (22) with $c_0 = (k - 1)/(2(k_1 - 1))$ and $C_0 = (k - 1)/\varepsilon$. Since $g \in [2\varepsilon, k_1 - 1]$, we have $gc_0 - (k - 1) \leq -(k - 1)/2$ and $gC_0 - (k - 1) \geq k - 1$. Let $f(z)$ be the logarithm of the right side of (22) without its error factor. Then $f'(z) = g + (\eta + 2 \sum_j \psi(x_j))/z + \mathcal{E}'(z)$, since $x'_j = 2x_j/z$. Also $\mathcal{E}'(z) = O(L^{-2})$, because $gs_a = O(L^{-2})$, $0 \leq \psi' \leq 1$ and the x_j are bounded. So $f' \geq g/2 \geq \varepsilon$ on $[c_0 L, C_0 L]$ for large N . Hence $f(c_0 L) = (gc_0 - (k - 1))L + O(\log L)$ is negative and $f(C_0 L) = (gC_0 - (k - 1))L + O(\log L)$ is positive, and f has exactly one zero ζ_N in the interval. By (22) the same signs hold for $\log S_{(n,a)}(z/N)$. The polynomial $S_{(n,a)}$ has nonnegative coefficients, zero constant term and the positive coefficient $k!$ at u^{k-1} , so the crossing is unique and lies in the interval. There $|f(Nu_{n,a})| = O(L^{-2} + L^2/N)$, and $f' \geq \varepsilon$ gives (23).

Step 8. The explicit expansion. For (24), take $f(\zeta_N) = 0$ and use $\mathcal{E} = O(L^{-1})$. This gives

$$g\zeta_N + \eta \log \zeta_N = (k - 1)L - \log(D_{k_1}(\beta)A^{2\ell}) - \sum_j \log \Phi(A^2\rho_j(\zeta_N/L)^2) + O(L^{-1}).$$

First $\zeta_N = c_* L + O(\log L)$. The derivative of each $\log \Phi(A^2\rho_j(z/L)^2)$ in z is $2\psi/z = O(L^{-1})$, so the sum of logarithms differs from its value at $\zeta_N = c_* L$ by $O(\log L/L)$. Substituting this into $\log \zeta_N$ and using (23) proves (24). $\square$

Remark C.2. The three terms of $\mathcal{E}$ have different sources. Replacing M by N in the exponent $A^2 Y$ gives a factor $\exp(-A^2 z \sum_j a_j/N)$, and the Gaussian average of the factors Φ gives back $\exp(z \sum_j a_j/N)$. Together they give the first term. The second term comes from the same Gaussian average, since the factors Φ and the power of w shift the saddle point. The last term does not depend on the sizes a_j , and for $\ell = 0$ and equal β_i it is $-(k - 1)/(8z)$, as in (6).

Take two equal large counts and one rare count, so that $g = 1$ and $\eta = \frac{5}{2}$, and let $z = \log N$. Then $x_1 = 2\rho$, and the first two terms of $z\mathcal{E}$ are $-\rho + 1.25\psi(2\rho)$, which nearly cancel at $\rho = 1$. So the first term alone is not the correction. The ratio of $S_{(n,a)}$ to the right side of (22) without $e^{\mathcal{E}}$ is 1.04 for $\rho = 1$ and $N = 2000$, and with the first term alone it would be 1.19. For $\rho = 8$ it is 0.36, 0.53, 0.63 and 0.70 at $N = 2 \cdot 10^3$, $2 \cdot 10^4$, $2 \cdot 10^5$ and $2 \cdot 10^6$, and with the factor $e^{\mathcal{E}}$ these become 0.52, 0.70, 0.79 and 0.85. For $\rho = 1$ it becomes 0.90. At $N = 2000$ we have $L = 7.6$, and the terms of order L^{-2} and L^2/N are comparable to γ_0/L , so this value does not improve. The explicit expansion (24) converges slowly. For $\rho = 8$ and $N = 10^6$ it gives 6.58, while $Nu_{n,a} = 14.99$ and $\zeta_N = 14.86$, since it expands around $z = c_* L$ with $c_* = 2$, while $Nu_{n,a}/L$ is only 1.08 (numerically verified, see Section 7).

Corollary C.3 (invisibility of subcritical rare counts). *Under the hypotheses of Theorem C.1, if every $a_j = o(N/L^2)$, the sum of the terms $\log \Phi$ in (24) is $o(1)$. So the sizes of the rare counts change the expansion only by $o(1)$. Their number ℓ still changes its leading and logarithmic coefficients.*

Proof. Since $\Phi(0) = 1$ and $0 \leq (\log \Phi)' \leq 1/2$ (Lemma A.1), we have $0 \leq \log \Phi(x) \leq x/2$, and here $x = A^2 c_*^2 \rho_j \rightarrow 0$. $\square$

The scale $N/(\log N)^2$ needs at least two coordinates of order N . It does not apply near a vertex, for example at an endpoint in the two-state case, where only one coordinate is large and $g = 0$ (Subsection 6.1). The theorem gives no error bound when some ρ_j is unbounded.

D Proofs near a vertex

We use the notation of Subsection 6.1 and prove Theorem 6.4. The first lemma collects what we need about the polynomial $\mathcal{H}$ of the rare counts, and the second writes $S_{(b,a)}$ exactly as an average of $\mathcal{H}$.

Lemma D.1 (elasticity and concavity of the rare-count polynomial). *The polynomial $\mathcal{H}$ is convex, $\log \mathcal{H}$ is concave on $(0, \infty)$, and κ is nondecreasing. Also*

$$\frac{1}{|a|+1} \leq y_a \leq 1, \quad \kappa(y) \leq \ell + \sqrt{\ell|a|y}, \quad \kappa(y) \geq \max\{\ell, \tfrac{1}{2}\sqrt{|a|y}\} \quad (0 < y \leq 1). \quad (27)$$

Proof. Step 1. One factor as a law. The polynomial $\mathcal{H}$ has nonnegative coefficients, so it is convex on $(0, \infty)$. For one factor B_a and $y > 0$, let J take values in $\{0, \dots, a-1\}$ with $\mathbb{P}(J=j)$ proportional to $\binom{a-1}{j} y^j / (j+1)!$. Since $B_a(y) = \sum_j y \binom{a-1}{j} y^j / (j+1)!$, the elasticity of B_a is $\kappa_a = yB'_a/B_a = \mathbb{E}[J+1]$. The ratio

$$\mathcal{R}(j) = \frac{(j+1)\mathbb{P}(J=j+1)}{\mathbb{P}(J=j)} = \frac{y(a-1-j)}{j+2}, \quad \mathcal{R}(a-1) = 0,$$

is nonincreasing in j . Summing the ratio identity gives $\mathbb{E}J = \mathbb{E}\mathcal{R}(J)$ and $\mathbb{E}J^2 = \mathbb{E}[(J+1)\mathcal{R}(J)]$. So

$$\text{Var } J = \mathbb{E}J + \text{Cov}(J, \mathcal{R}(J)) \leq \mathbb{E}J.$$

The covariance is nonpositive because, for an independent copy J' , twice the covariance equals $\mathbb{E}[(J-J')(\mathcal{R}(J) - \mathcal{R}(J'))] \leq 0$.

Step 2. Concavity and monotonicity. Since $y^2 B''_a / B_a = \mathbb{E}[J(J+1)]$, we get

$$(\log B_a)'' = \frac{\text{Var } J - \mathbb{E}[J+1]}{y^2} < 0, \quad y\kappa'_a = \text{Var } J \geq 0.$$

Summing over the factors proves the concavity of $\log \mathcal{H}$ and the monotonicity of $\kappa = \sum_i \kappa_{a_i}$.

Step 3. The elasticity bounds. The differential equation $yB''_a + yB'_a = aB_a$, which we check on the coefficients, gives

$$ay = \kappa_a^2 + \text{Var } J + (y-1)\kappa_a.$$

Using $0 \leq \text{Var } J \leq \kappa_a - 1$ we get

$$\kappa_a \leq 1 + \sqrt{ay}, \quad \kappa_a^2 + y\kappa_a \geq ay + 1.$$

For $0 < y \leq 1$ the second inequality gives $(\kappa_a + \frac{1}{2})^2 > ay$, and $\kappa_a \geq 1$. So $\kappa_a \geq \max\{1, \sqrt{ay} - 1/2\} \geq \frac{1}{2}\sqrt{ay}$. Summing over the factors proves the elasticity bounds, using Cauchy-Schwarz, $\sum_i \sqrt{a_i} \leq \sqrt{\ell|a|}$, for the upper bound and $\sqrt{\sum_i x_i} \leq \sum_i \sqrt{x_i}$ for the lower one.

Step 4. The root. Finally, $B_a(1) \geq 1$, from its term $m=1$, and Lemma 6.1 gives

$$B_a(y) \leq y\Phi(ay) \leq ye^{ay/2},$$

where $(j+1)! \geq 2^j$ proves the last inequality term by term. So $\mathcal{H}(1) \geq 1$, while, since $|a|+1 \geq 2$, $\mathcal{H}(1/(|a|+1)) \leq (|a|+1)^{-\ell} e^{|a|/(2(|a|+1))} \leq \frac{1}{2}e^{1/2} < 1$. $\square$

Lemma D.2 (an exact compound-binomial representation). *For $0 < u < 1$, put $v = u/(1-u)$. Let Γ_2 have the gamma distribution with shape 2 and rate 1, let $E_1, \dots, E_{b-1}$ be independent rate-one exponential variables, and let $Z_1, \dots, Z_{b-1}$ be independent Bernoulli variables with parameter u . Take all these variables independent and set*

$$\mathcal{X} = \Gamma_2 + \sum_{j=1}^{b-1} Z_j E_j.$$

Then

$$S_{(b,a)}(u) = (1-u)^{|a|} \mathbb{E}[\mathcal{H}(v\mathcal{X})], \quad \mathbb{E}e^{s\mathcal{X}} = (1-s)^{-2} \left(1 + \frac{us}{1-s}\right)^{b-1} \quad (s < 1). \quad (28)$$

In particular, set $y = bu^2$, $\kappa = \kappa(y)$ and $s = \kappa/[bu(1-u)]$. Whenever $s < 1$,

$$(1-u)^{|a|} \mathcal{H}(y) \leq S_{(b,a)}(u), \quad (29)$$

$$S_{(b,a)}(u) \leq (1-u)^{|a|} \mathcal{H}(y) e^{-\kappa} (1-s)^{-2} \left(1 + \frac{us}{1-s}\right)^{b-1}. \quad (30)$$

Proof. By (21) with $N = b + |a|$,

$$S_{(b,a)}(u) = \frac{(1-u)^{b+|a|-1}}{v} \int_0^\infty e^{-t} B_b(vt) \mathcal{H}(vt) dt.$$

The integral of $e^{-t} B_b(vt)$ is $\sum_m \binom{b-1}{m-1} v^m = v(1+v)^{b-1}$. After normalization its density is $\sum_{j=0}^{b-1} \binom{b-1}{j} u^j (1-u)^{b-1-j} t^{j+1} e^{-t} / (j+1)!$, a mixture of gamma densities with shape $J+2$, where $J \sim \text{Bin}(b-1, u)$. This is exactly the distribution of $\mathcal{X}$, which proves the identity and its moment generating function. Since $\mathbb{E}\mathcal{X} = 2 + (b-1)u$, we have $v\mathbb{E}\mathcal{X} - bu^2 = u(2-u+bu^2)/(1-u) > 0$, so $v\mathbb{E}\mathcal{X} \geq y$. Jensen's inequality for the convex function $\mathcal{H}$, and the monotonicity of $\mathcal{H}$, prove (29). Concavity of $\log \mathcal{H}$ gives, for $w > 0$,

$$\mathcal{H}(w) \leq \mathcal{H}(y) \exp\left(\frac{\kappa}{y}(w-y)\right).$$

Insert $w = v\mathcal{X}$, note that $\kappa v/y = s$, and use the moment generating function to prove (30). $\square$

Proof of Theorem 6.4. Write $S = S_{(b,a)}$. Put $u_0 = \sqrt{y_a/b}$ and $\kappa_0 = \kappa(y_a)$. Since S increases on $u > 0$, it suffices to show $\log S(u_+) > 0$ and $\log S(u_-) < 0$ at $u_\pm = (1 \pm \delta)u_0$. Note that $\delta \leq 1/2$ forces $\varepsilon \leq 1/(32c_\ell) \leq 1/96$, and that $u_0 \leq \sqrt{1/b} \leq \varepsilon$ since $y_a \leq 1$.

Step 1. The upper point. We have $u_+ \leq \frac{3}{2}\varepsilon < 1/4$. Since $\log \mathcal{H}(y_a) = 0$ and $(\log \mathcal{H})' = \kappa/y$, monotonicity of κ gives $\log \mathcal{H}(bu_+^2) \geq \kappa_0 \log(bu_+^2/y_a) = 2\kappa_0 \log(1+\delta)$. The lower bound in Lemma D.1 gives $|a|u_0/\kappa_0 \leq 2\sqrt{|a|/b} \leq 2\varepsilon$. Hence, by (29) and $-\log(1-x) \leq x/(1-x)$,

$$\log S(u_+) \geq 2\kappa_0 \log(1+\delta) - \frac{|a|u_+}{1-u_+} \geq \kappa_0 \left(\frac{4}{3}\delta - 4\varepsilon\right) > 0.$$

Here we used $2\log(1+\delta) \geq \frac{4}{3}\delta$ for $\delta \leq \frac{1}{2}$, and $|a|u_+/(1-u_+) \leq \frac{4}{3} \cdot \frac{3}{2}|a|u_0 \leq 4\kappa_0\varepsilon$. The last inequality holds since $\frac{4}{3}\delta = \frac{64}{3}c_\ell\varepsilon > 4\varepsilon$.

Step 2. The lower point. Write $y = bu_-^2$ and $\kappa = \kappa(y)$. The inequalities $y_a \geq 1/(|a| + 1)$, $\delta \leq 1/2$ and the upper bound in (27) imply

$$u_- \leq \varepsilon, \quad bu_- \geq \frac{1}{2\varepsilon}, \quad s = \frac{\kappa}{bu_-(1-u_-)} \leq 2(2\ell + \sqrt{\ell})\varepsilon \leq 4c_\ell\varepsilon \leq \frac{1}{8}.$$

For the bound on s , note that $\kappa/(bu_-) \leq \ell/(bu_-) + \sqrt{\ell|a|y}/(bu_-) \leq 2\ell\varepsilon + \sqrt{\ell|a|/b} \leq (2\ell + \sqrt{\ell})\varepsilon$, and $1 - u_- \geq \frac{1}{2}$. We use $\log(1+x) \leq x$, $-\log(1-s) \leq s/(1-s)$ and $u_-, s \leq 1/4$ in (30), and we drop the nonpositive term $|a|\log(1-u_-)$. This gives

$$\begin{aligned} \log \frac{S(u_-)}{\mathcal{H}(y)} &\leq -\kappa + \frac{2s}{1-s} + \frac{bu_-s}{1-s} \\ &= \kappa \left(\frac{1}{(1-u_-)(1-s)} - 1 \right) + \frac{2s}{1-s} \\ &\leq 5\kappa(u_- + s) \leq 25c_\ell\kappa\varepsilon. \end{aligned}$$

Here the equality uses $bu_-s = \kappa/(1-u_-)$. The next inequality uses $(1-u_-)(1-s) \geq \frac{21}{32}$, so the first term is at most $\frac{32}{21}\kappa(u_- + s)$, and $\kappa \geq \ell \geq 1$, so the second is at most $\frac{16}{7}\kappa s$. The last one uses $u_- + s \leq 5c_\ell\varepsilon$. Again by monotonicity of κ ,

$$\log \mathcal{H}(y) = - \int_y^{y_a} \frac{\kappa(x)}{x} dx \leq 2\kappa \log(1-\delta) \leq -2\kappa\delta = -32c_\ell\kappa\varepsilon.$$

Hence $\log S(u_-) \leq \kappa(-32c_\ell + 25c_\ell)\varepsilon < 0$. Since S is increasing on $u > 0$, the two strict inequalities bracket the crossing. $\square$

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Conflict of interest. The author has no conflict of interest and no competing interests to declare.

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